



Systemic Pathology Lab Final Summary

Dr. Hamzah AD.

Cardio

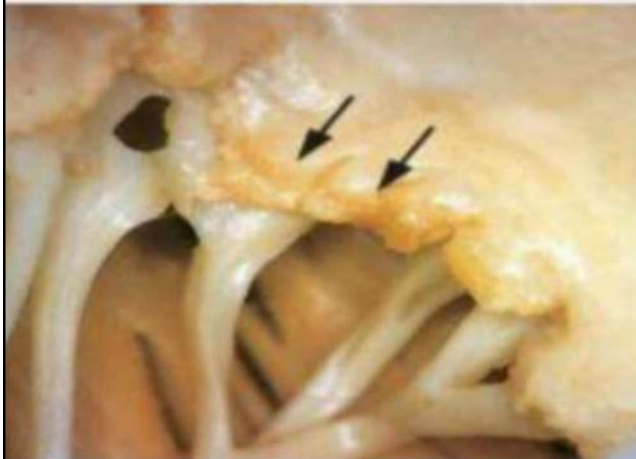
Coarctation of the aorta



segmental narrowing of aorta

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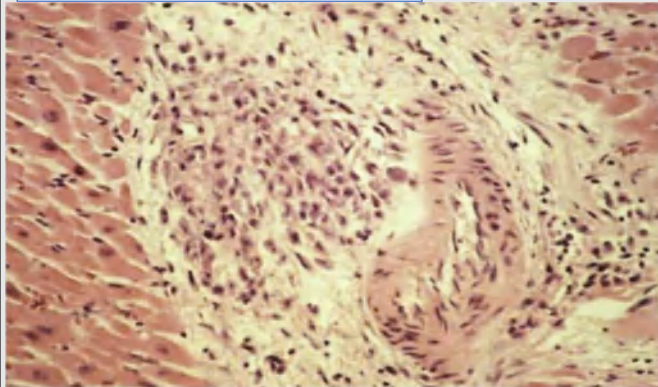
Rheumatic heart disease



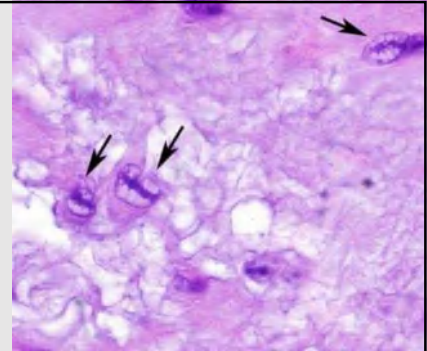
Aschoff body (inflammatory nodules) in acute rheumatic carditis

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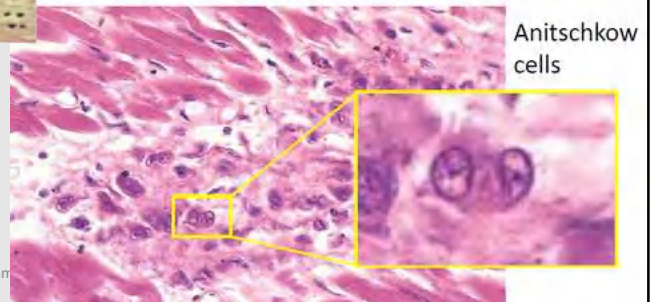
Rheumatic heart disease



Aschoff body in acute rheumatic carditis



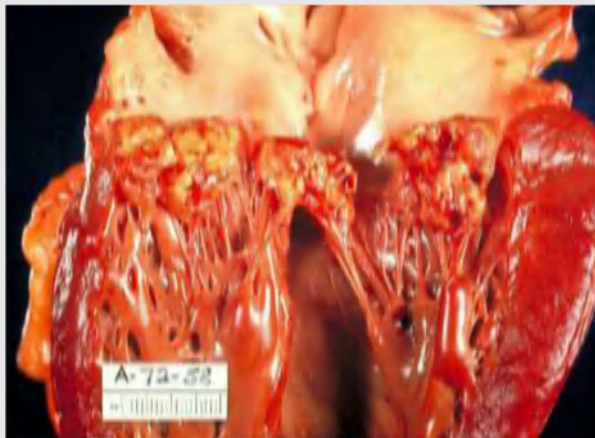
Binucleated macrophage



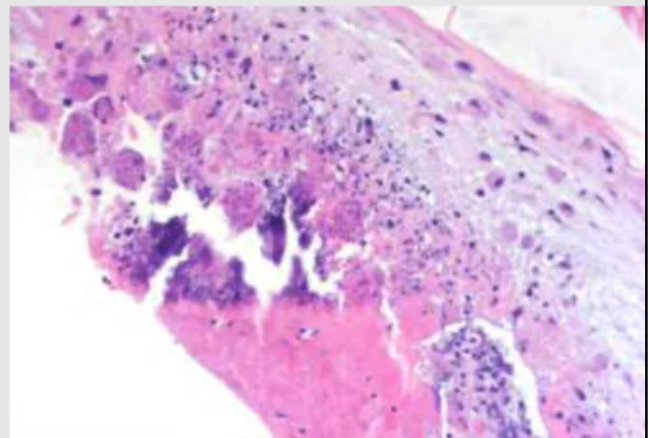
Anitschkow cells

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Infective endocarditis



Mitral Valve vegetation



bacterial colonies + fibrin deposition + inflammatory cells

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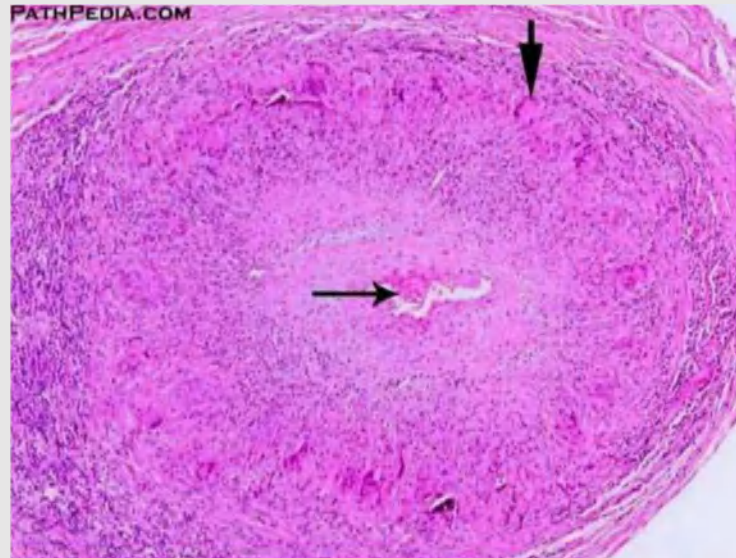
Temporal Giant cell arteritis or Takayasu vasculitis

The general features of vasculitis

- thickening of the wall
- narrowing of the lumen
- leukocyte infiltrate

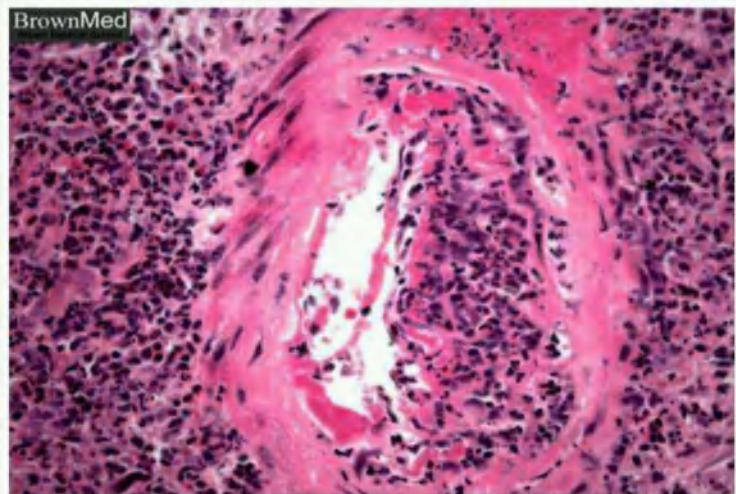
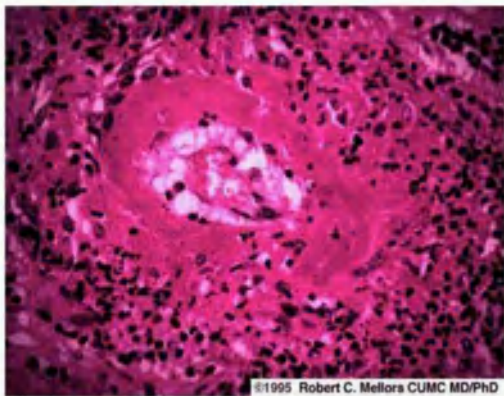
Characteristic Features:

- granulomatous inflammation
- giant cells



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Polyarteritis nodosa or Kwasaki

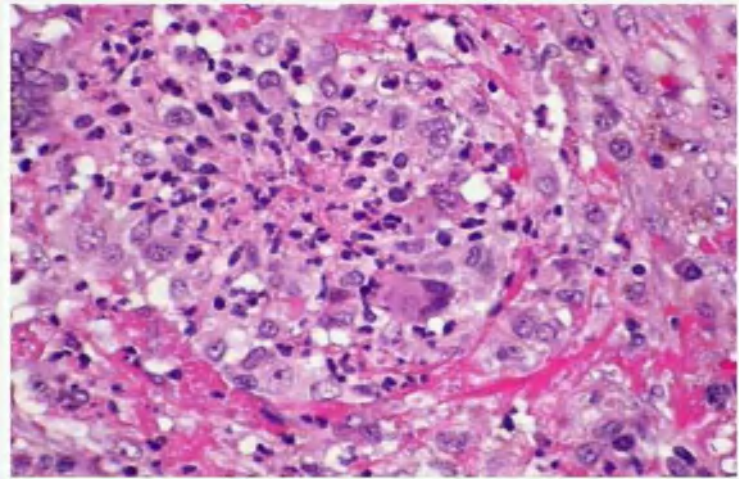


Characteristic Features:

- general features of vasculitis observed
- necrotizing vasculitis (Fibrinoid necrosis).

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Wegener's disease



Characteristic Features:

- necrotizing granuloma (Epithelioid cells and neutrophil at the middle).
- We can't see the blood vessel wall, there's destruction of the wall.
- Invasion to the surrounding tissue.

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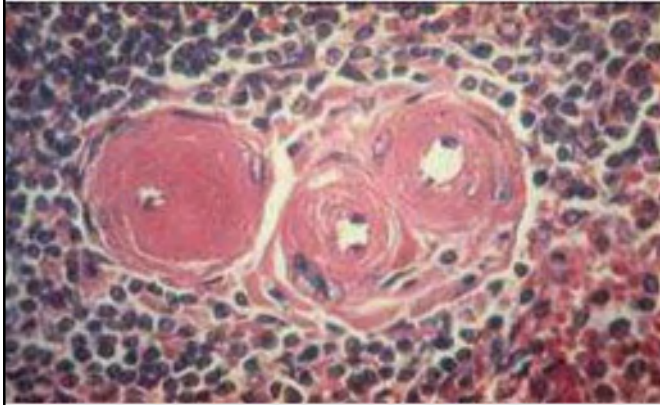
Abdominal Aortic aneurysm



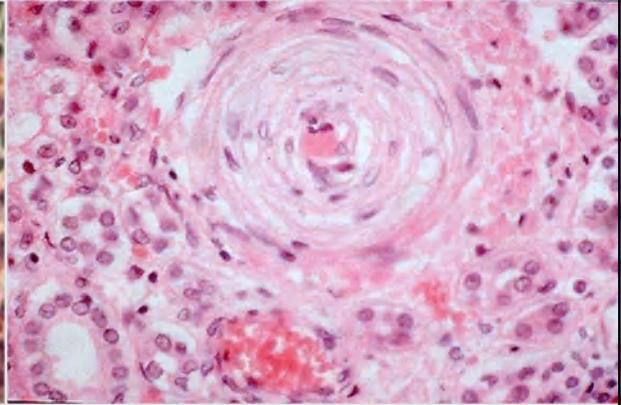
Abnormal permanent segmental dilatation of blood vessels

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Arteriolosclerosis (Hypertension)



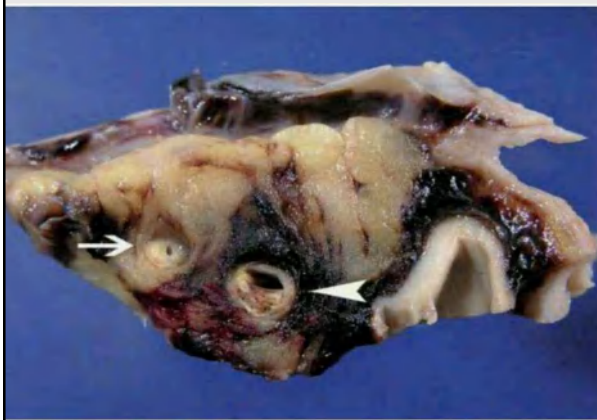
Hyaline arteriolosclerosis
• Amorphous material.



hyperplastic arteriolosclerosis (Onion shaped)

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Atherosclerosis



Gross picture of Atherosclerosis shows mass in the lumen



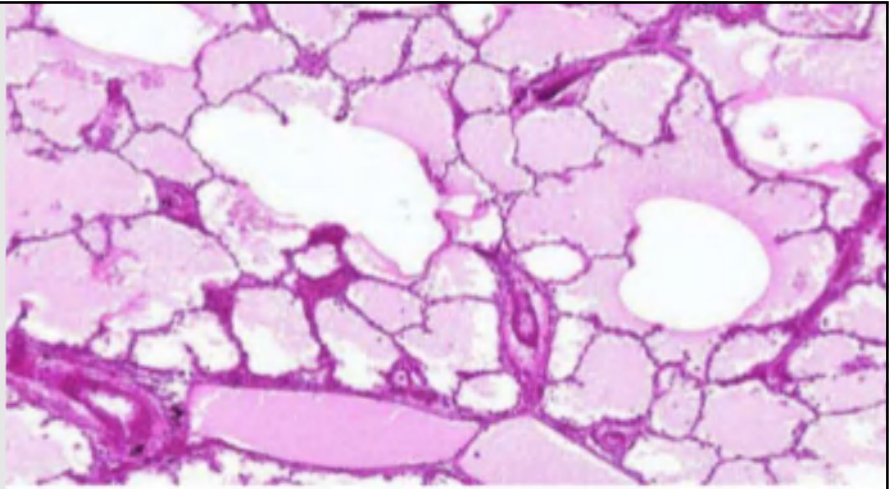
Characteristic Features:

- Atheroma composed of:
 - ✓ fibrous cap
 - ✓ fibrin
 - ✓ leukocyte
 - ✓ smooth muscle
 - ✓ fat (cholesterol)
- all are found in intima**

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Respiratory System

Pulmonary Edema



Diagnosis:

Pulmonary edema (Mild)

Characteristic Features:

Normal alveolar wall.

Accumulation of fluids in the alveolar space

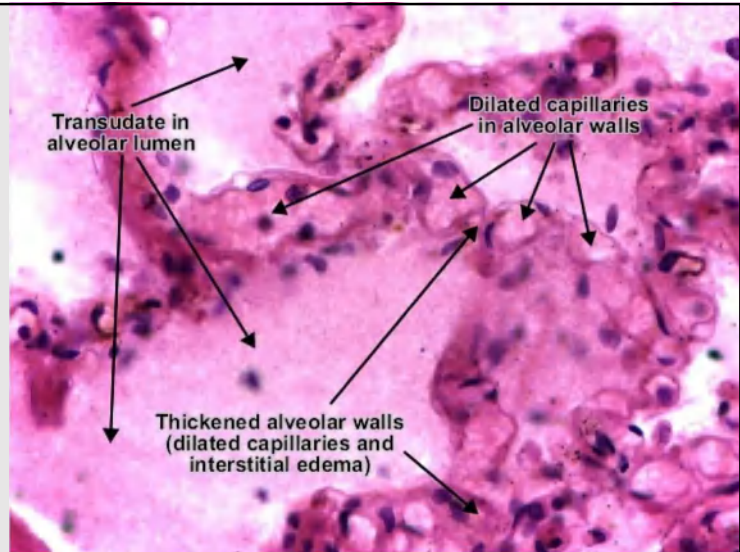
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Pulmonary Edema

pulmonary edema (Severe)

Characteristic Features:

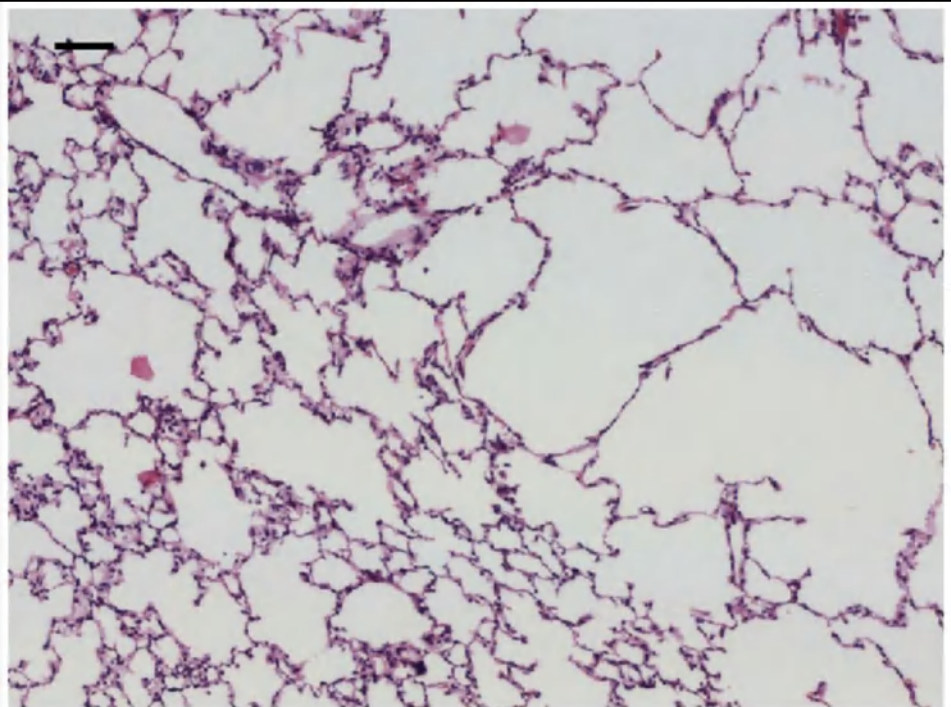
- Transudate in the alveolar lumen
- Dilated capillaries in the alveolar wall
- Thickened alveolar walls (dilated capillaries and interstitial edema)



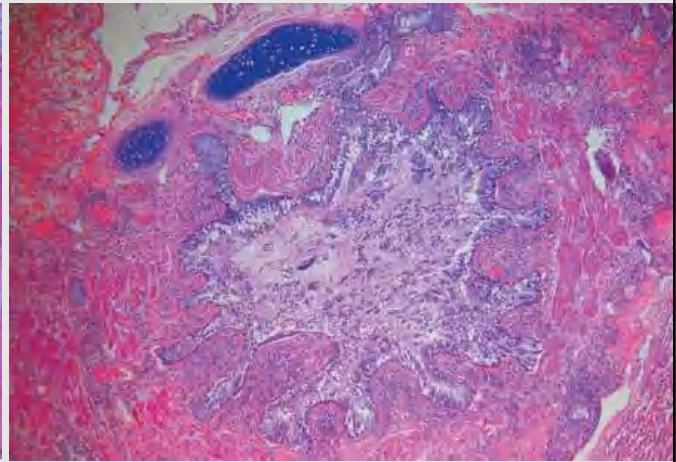
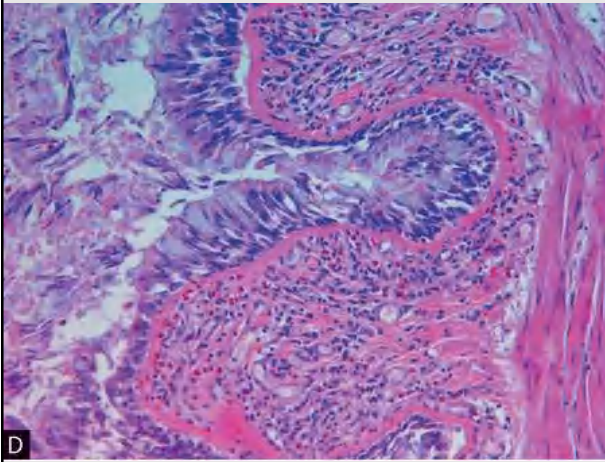
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Emphysema

- Dilated alveoli
- Thin walls
- without fibrosis



Asthma

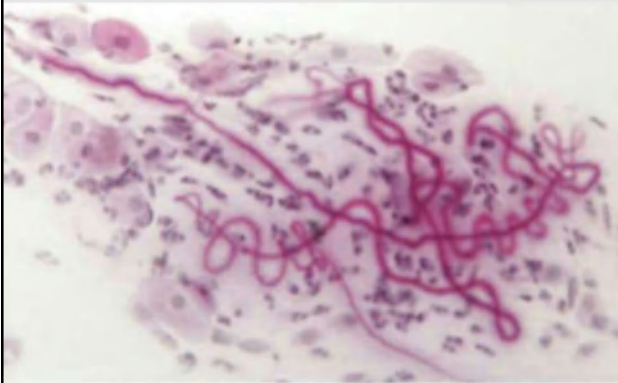


Characteristic Features:

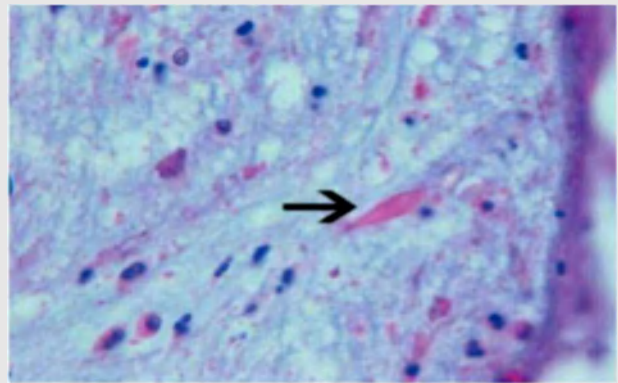
- 1-Hyperplasia and hypertrophy of the Smooth Muscles.
- 2-Features of inflammation: (edema, neutrophil infiltrate, eosinophilic infiltrate)
- 3-Hyperplasia of mucus glands.

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Asthma



Cushman spirals:
microscopic findings in the sputum of the asthmatics. They are spiral-shaped mucus plugs from sub-epithelial mucus gland ducts of bronchi.



Charcot-Leyden crystals:
Microscopic crystals composed of eosinophil protein galectin-10 found in people who have allergic diseases.
(morphologic hallmark of eosinophil-related disease)

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Acute pneumonia



Red hepatization of the lung

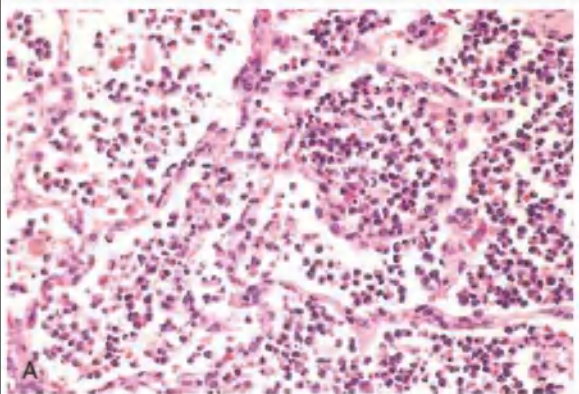
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Chronic pneumonia



Gray hepatization of the lung

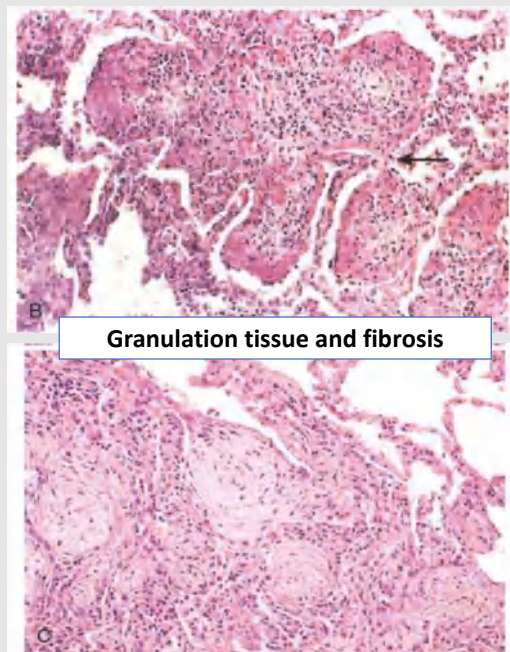
Acute bacterial pneumonia



Inflammatory infiltrate with neutrophils in the alveolar wall and intra-alveolar space.

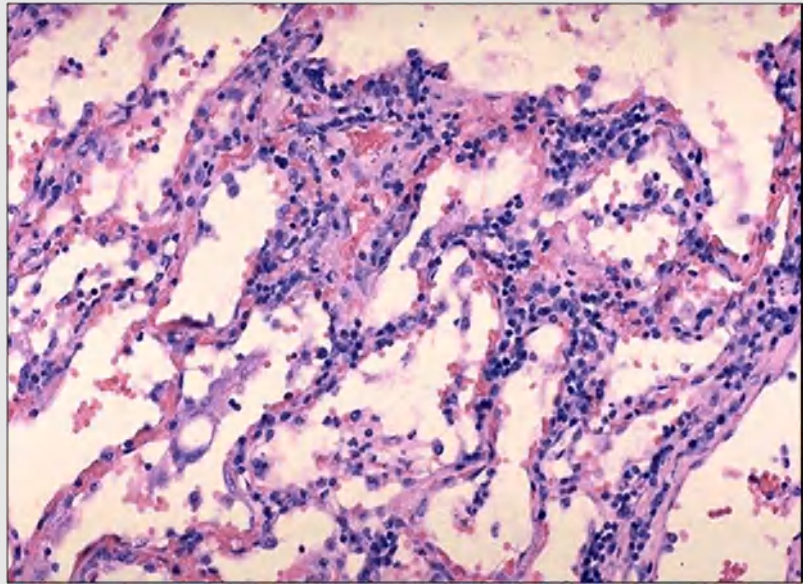
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Chronic pneumonia



Granulation tissue and fibrosis

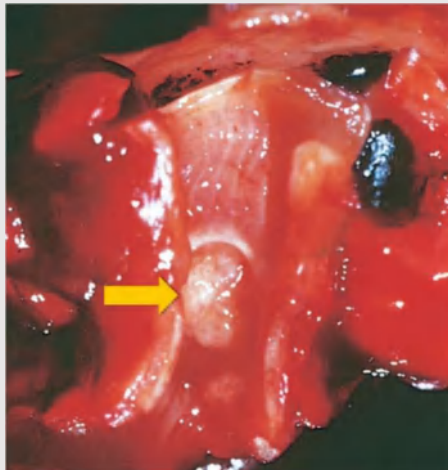
Atypical Pneumonia



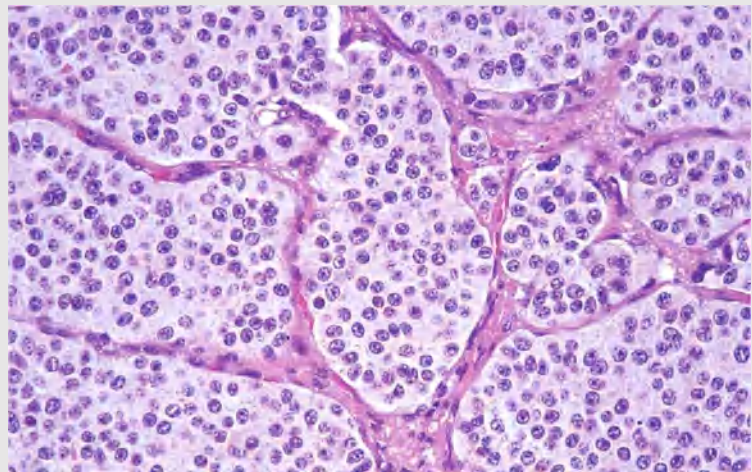
Inflammatory cell infiltrate (lymphocyte and plasma cells) in the interstitial lung tissue.

>> Thickening of the wall.

Lung Carcinoid tumor



2 – 4 cm yellow-red mass
White to tan in cross section
No necrosis
No hemorrhage



Nests of round cells separated by fibrous tissue
with Salt and Pepper chromatin pattern of the cells.

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Thyroid

Colloid goiter



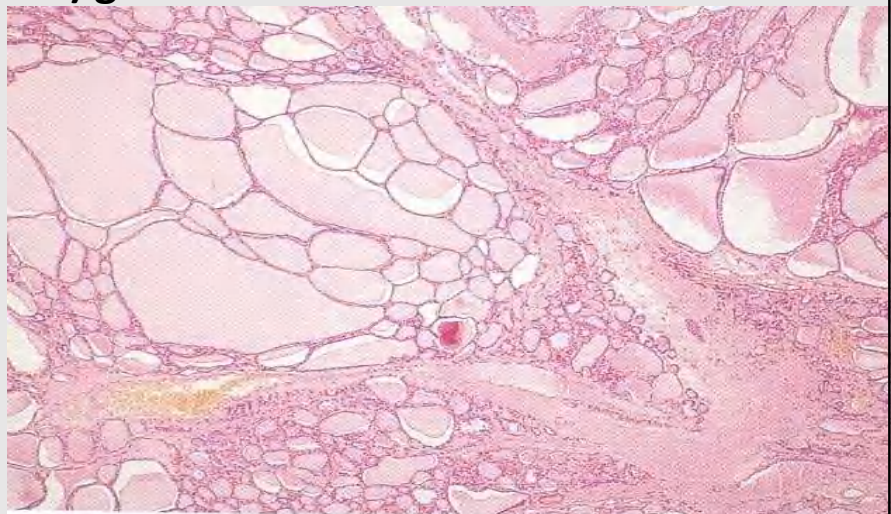
Enlarged follicles lined by low cuboidal (flattened) cells, Distended with colloid.

Nodular (multinodular) goiter



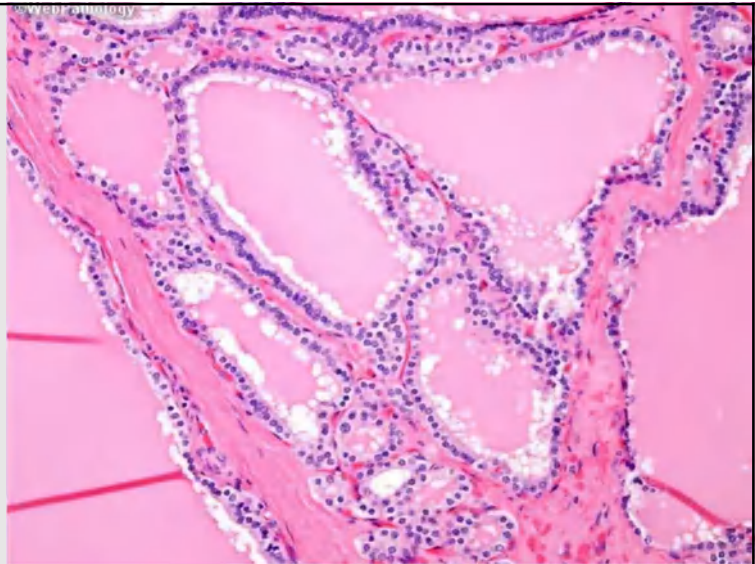
Cut-section shows variable sized nodules separated by fibrous tissue, Some nodules are solid (hyperplastic) other are cystic containing brownish colloid. Areas of hemorrhage, calcification and fibrosis are present.

Nodular (multinodular) goiter



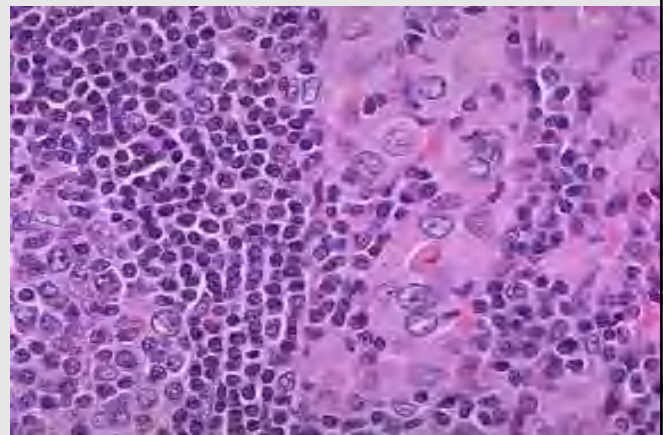
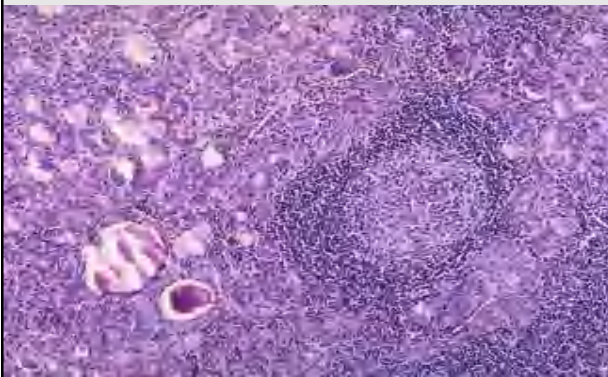
Some follicles are normal (cubical cells). Some follicles are hyperplastic (columnar cells).
Some follicles show colloid distension.
Some follicles may rupture → hemorrhage, fibrosis and calcification.

GRAVIS DISEASE



- Hyperplastic follicles lined by tall columnar cells and more crowded.
- Formation of small papillae.
- The colloid is pale with scalloped margins (vacuolated); due to rapid absorption.
- Vascular stroma containing several lymphocytes that may form aggregates.

Chronic Lymphocytic (Hashimoto's) thyroiditis



Diffuse infiltration by mononuclear inflammatory cell (small lymphocyte, plasma cells), and well developed germinal centers.

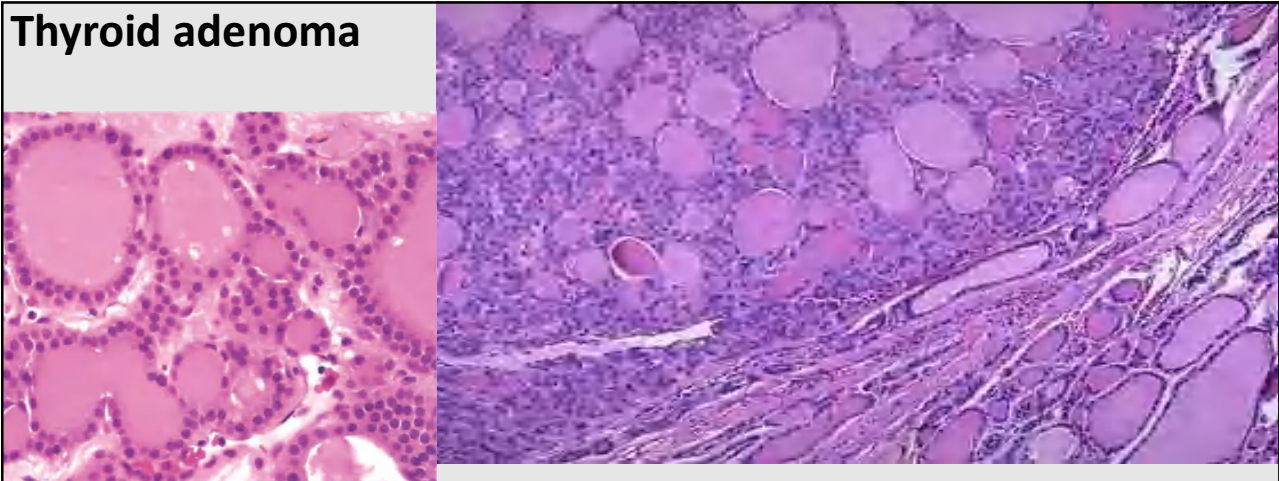
Atrophy of the thyroid follicles that appear small and may lined by epithelial cells that have abundant eosinophilic granular cytoplasm and called **hurthle** or **oxyphil** cells.

Thyroid adenomas



Solitary., Encapsulated and well-demarcated from the surrounding normal thyroid.
Cut-section → gray white to red-brown depending on the amount of colloid.
Areas of hemorrhage, necrosis and calcification may appear.

Thyroid adenoma



Well-Defined capsule demarcating the tumor cells from normal thyroid.
Usually normally appearing follicles containing colloid.
May be micro-follicular, or macro-follicular.
Papillary change is not a typical feature, and if present increase the incidence of papillary carcinoma.

Papillary carcinoma



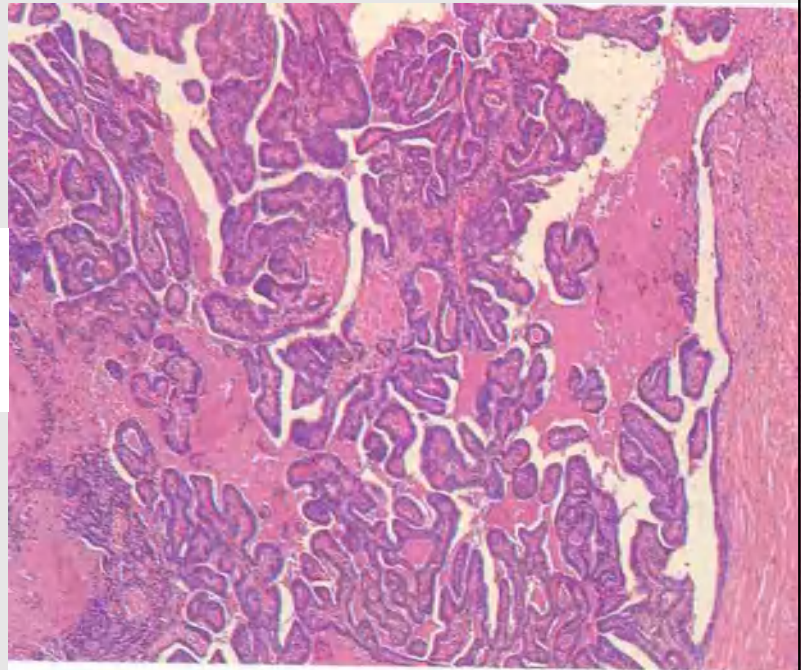
Solitary or multifocal lesion. May be:

- well circumscribed and even encapsulated, Or
- ill-defined and infiltrative.

May shows area of fibrosis and calcification and often cyst.

Cut-section → granular → sometimes grossly papillary foci.

Papillary carcinoma

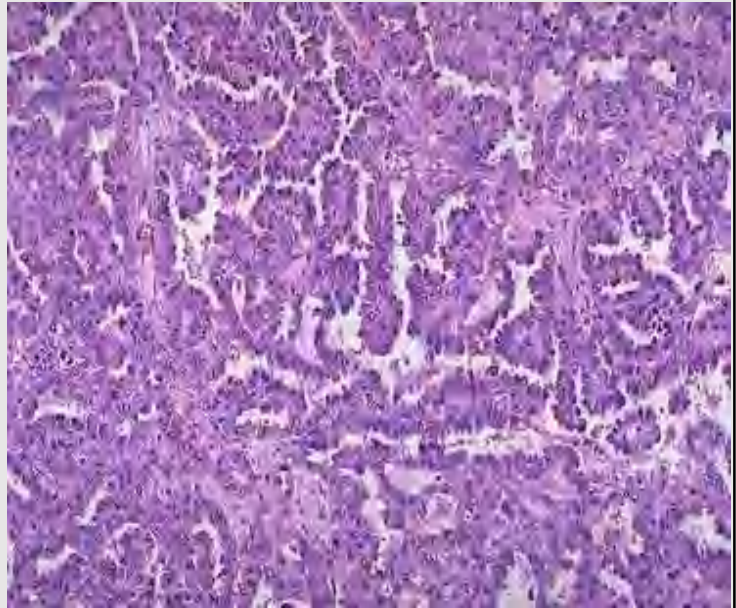


Well-formed papillary projections covered with a single layer of cuboidal epithelium.

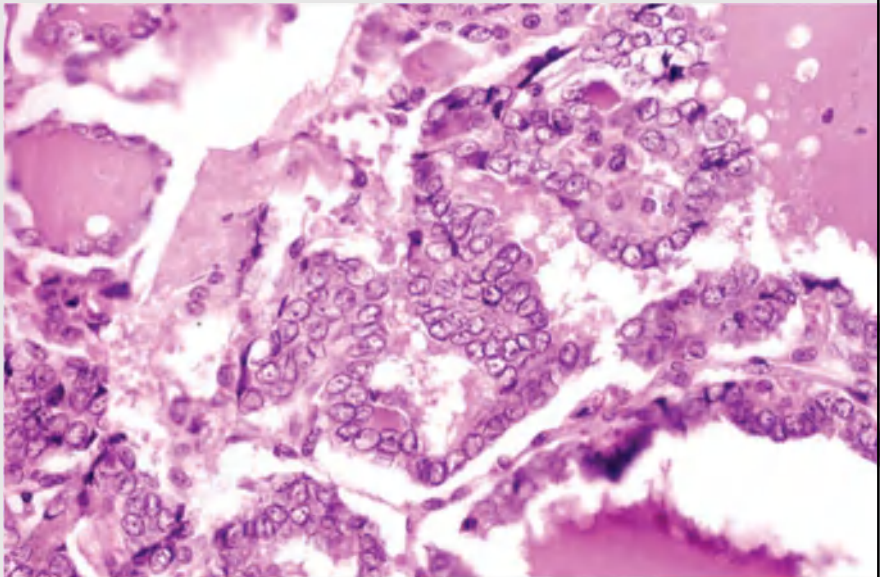
The papillae have a core of vascular connective tissue.

Papillary carcinoma

- Well-formed papillary projections covered with a single layer of cuboidal epithelium.
- The papillae have a core of vascular connective tissue.
- The nuclei are of uniform size, pale stained, and show mitotic features.
- May show calcified areas that appear as dense bluish area and are called psammoma bodies.
- May show foci of lymphocytic infiltration.



Papillary carcinoma



well-formed papillae are lined by cells with characteristic **empty-appearing nuclei**, sometimes termed ***Orphan Annie eye***.

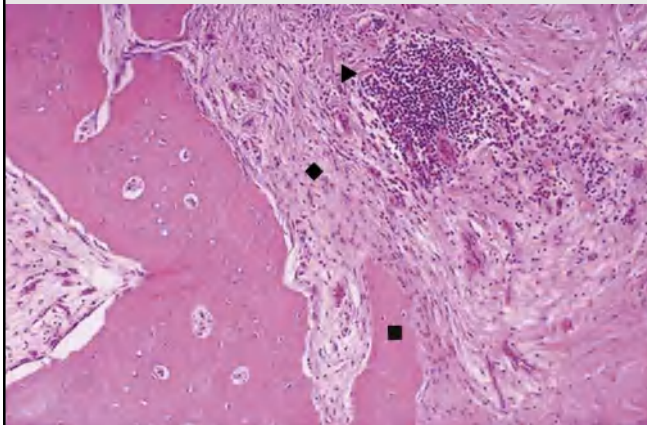
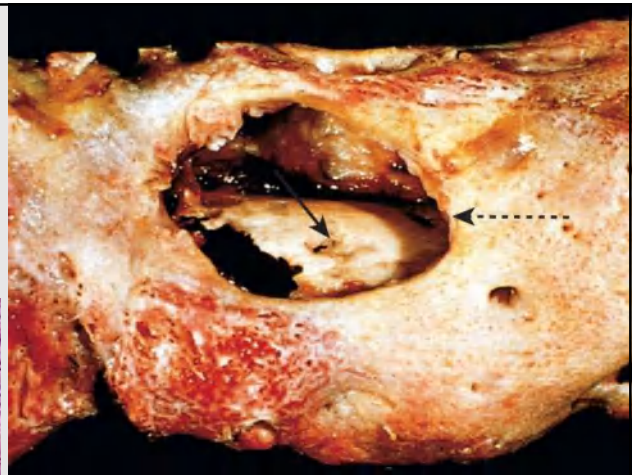
Bone

Chronic osteomyelitis

Gross:

The ***solid arrow*** points to necrotic bone in the center of a draining abscess (**sequestrum**).

The ***interrupted arrow*** is a rim of new bone formation (**involucrum**).



Microscopic:

- Shown here in the marrow is fibrosis ().
- Chronic inflammatory cell infiltrates (▶).
- The bony trabeculae have become disorganized, and the bone is devitalized ().

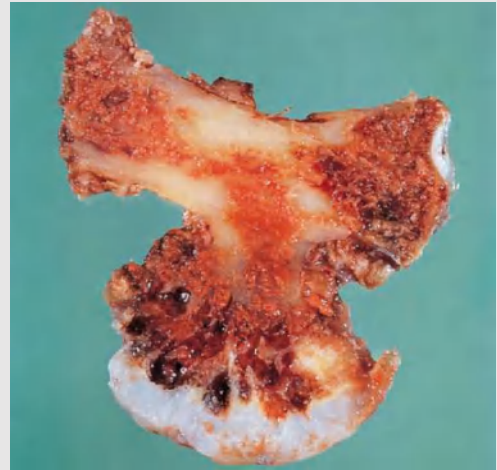
Osteochondroma

- A slowly growing solitary mass.
- Extends outward from the metaphyseal region of a long bone (**exostosis**).
- Arising most often **before growth plate closure**.
- The **knee** is the most common site.
- Malignant transformation is **uncommon** in solitary lesions.



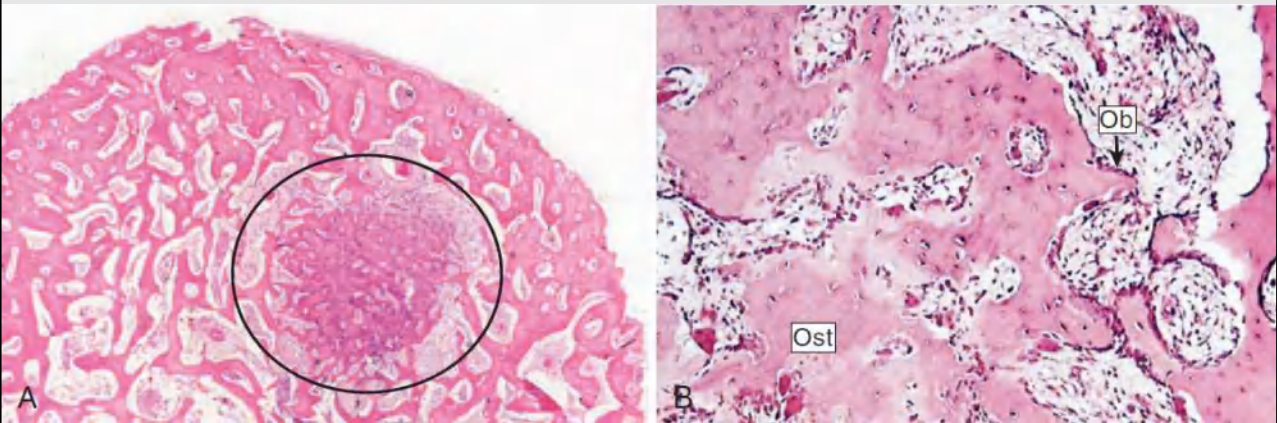
- A benign cartilaginous cap on the left.
- Underlying bony cortex on the right.

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Osteochondroma of the femur showing a bluish-white cartilaginous cap overlying a bony cortex.

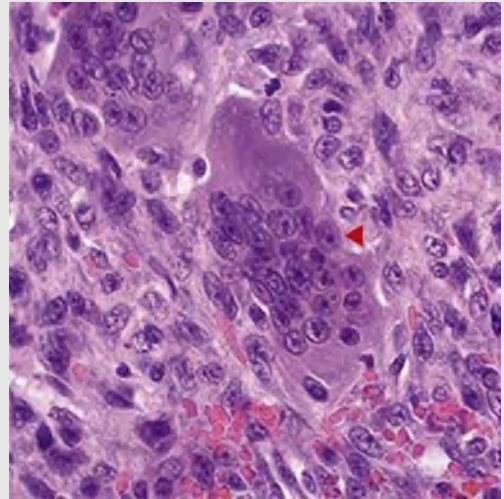
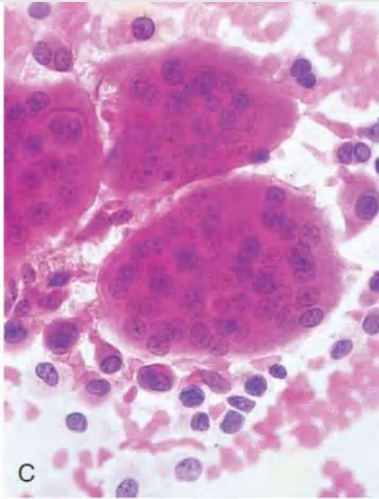
Osteoid osteoma



- A**, typical osteoid osteoma in the medullary bone of the tibia (*circle*).
- B**, At higher magnification, the tumor consists of irregular masses of recently formed osteoid (Ost) produced by numerous active but benign osteoblasts (Ob).

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Giant cell tumor

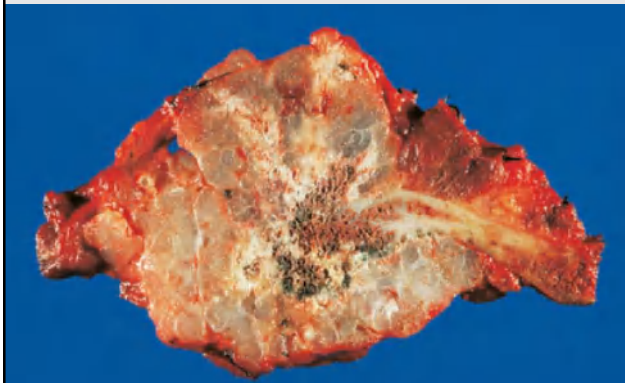


Giant cell tumor of bone

B, Gross showing a hemorrhagic mass at the distal end of the femur arising within epiphyseal region and extending into the metaphysis. It has destroyed the cortex and extended into the adjacent tissue.

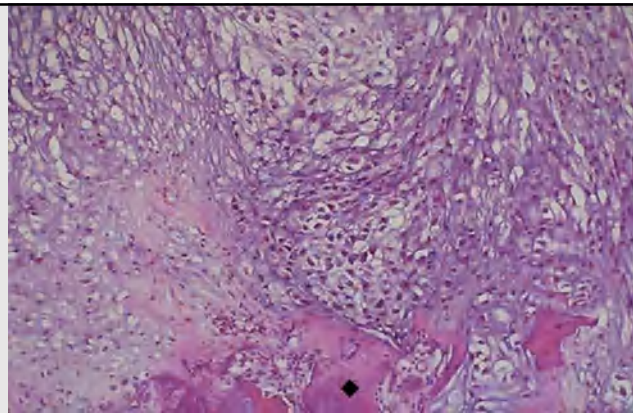
C, Microscopic image Giant cell tumors of bone are composed of osteoclast-like multinucleated giant cells in a sea of round-to-oval mononuclear stromal cells.

Chondrosarcoma



Chondrosarcoma.

Gelatinous-appearing gray-tan tumor infiltrating the bone.



Chondrosarcoma, microscopic

The tissue is still recognizable as cartilage, and there are chondrocytes within clear lacunae, surrounded by a bluish matrix, but there is no orderly pattern, and there is increased cellularity and atypia.

At the bottom the chondrosarcoma is invading and destroying bone.

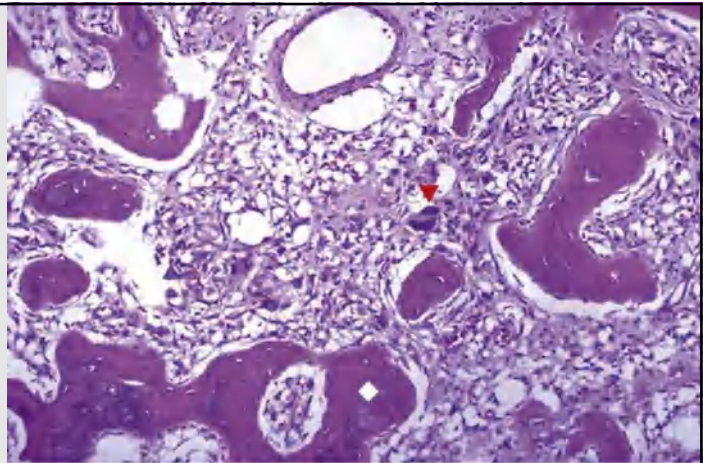
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Osteosarcoma



Osteosarcoma, gross

- Irregular mass lesion arising within the metaphysis at the upper tibia in this section.
- It breaks through the bone cortex and extends into adjacent soft tissue.
- The tumor tissue is firm and tan-white.

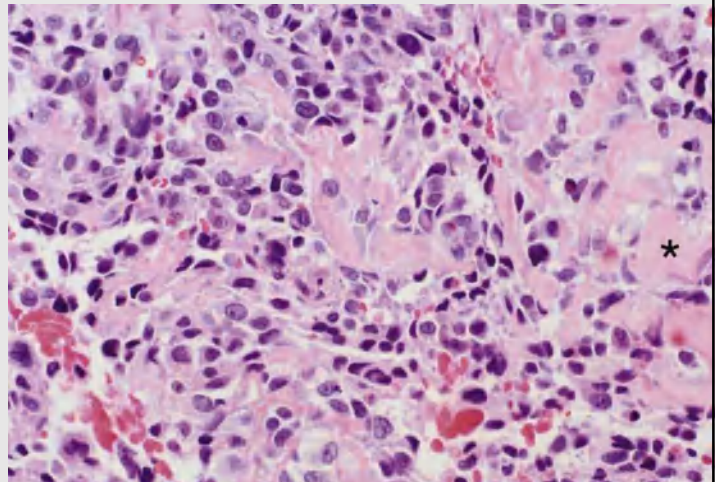


- Very pleomorphic cells, many with a spindle shape.
- One Large, bizarre multinucleated cell (**red arrow**) with very large nuclei is present.
- Nuclear hyperchromatism.
- Islands (**white arrow**) of reactive new woven bone forming.

Osteosarcoma

Osteosarcoma, microscopic

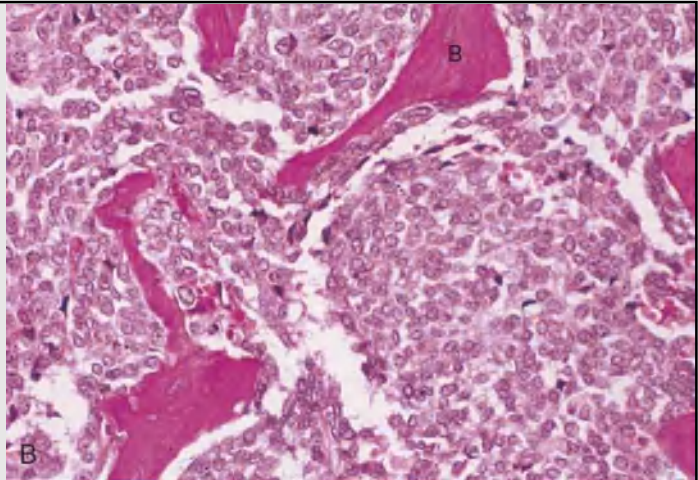
- The neoplastic pleomorphic cells of osteosarcoma are shown to be making pink osteoid (*).
- Osteoid production by a sarcoma is **diagnostic** of an osteosarcoma.
- Additional microscopic elements of an osteosarcoma include:
Vascular proliferation,
cartilaginous matrix,
and fibrous connective tissue.



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Ewing Sarcoma

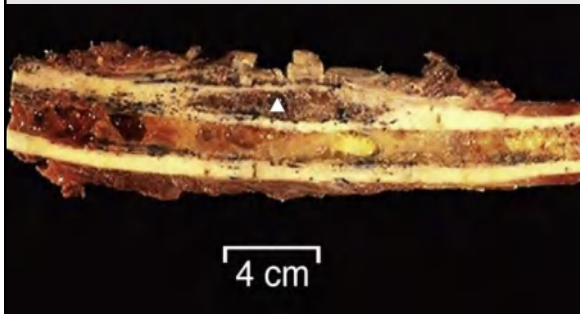
- This primary bone tumor mainly arises within the **medullary cavity in the diaphysis of long bones** and pelvis.
- Mainly in the first two decades of life
- Slight male predominance.



- A**, Ewing sarcoma originates in the **diaphysis** of long bones.
B, Histologically, the tumor is composed of a **uniform** population of small blue cells infiltrating the spaces between the bone trabeculae.

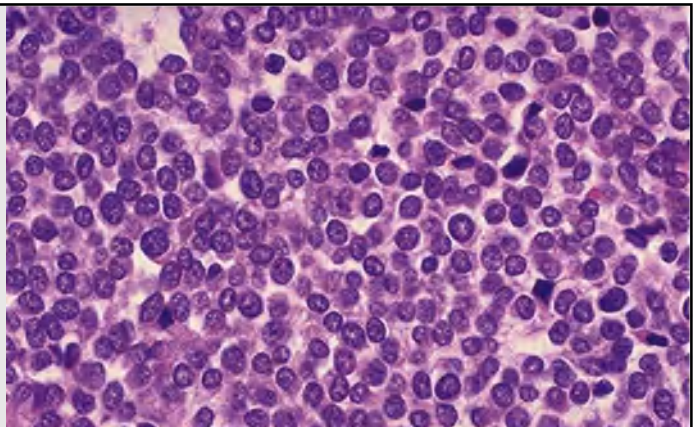
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Ewing Sarcoma



Ewing sarcoma, gross

The fibular tumor mass shown here is breaking through (*arrow*) the cortex. The **tan tumor** tissue has prominent areas of **reddish hemorrhage** and **brownish necrosis**.



Ewing sarcoma, microscopic

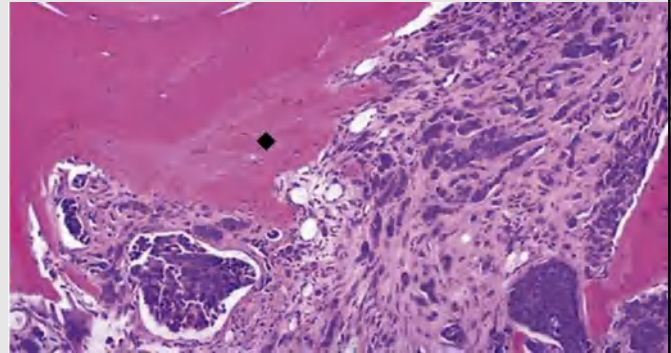
- A small round blue cell tumor of childhood.
- Marked cellularity and the high n/c ratio of these tumor cells, which are larger than lymphocytes.
- There is little intervening stroma.
- Mitoses are present.
- PAS stain may reveal abundant glycogen within tumor cell cytoplasm.

Bone Metastasis



Metastases, gross

This sagittal section of vertebral bone at autopsy shows multiple foci of pale and irregular metastatic tumor.



Metastases, microscopic

Metastatic ductal carcinoma of breast appears within vertebral bone, and it is filling the marrow cavity. There is reactive new bone at the margin of the carcinoma.

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GIT

PAROTID GLAND PLEOMORPHIC ADENOMA (MIXED TUMOR) - Gross

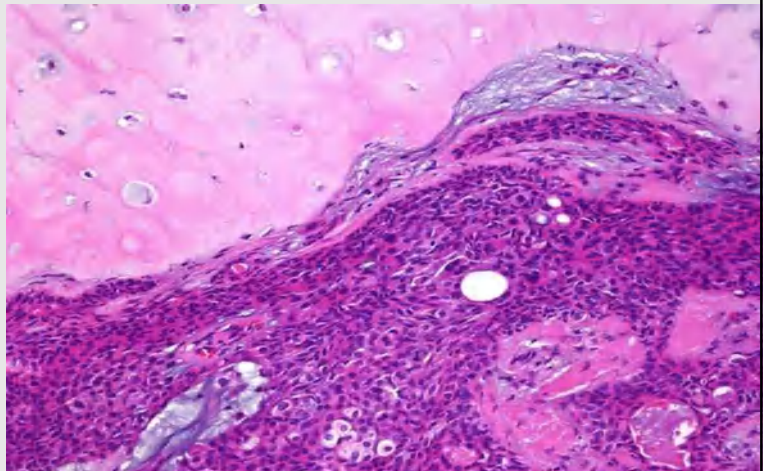
- ❖ Generally benign
- ❖ Have BOTH connective tissue (i.e., usually cartilaginous) components as well as glandular components, hence the name pleomorphic or mixed.
- ❖ They generally look and feel like little round soft cartilage balls.



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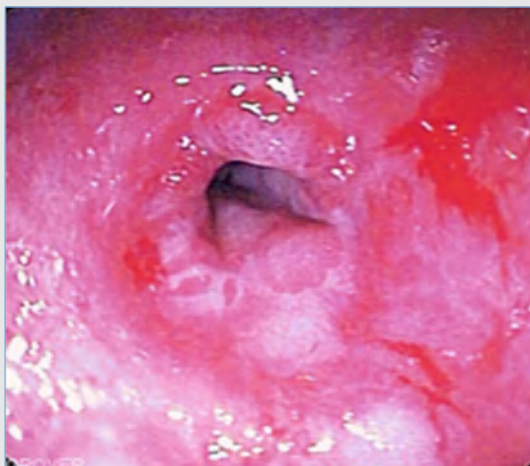
PLEOMORPHIC ADENOMA - Microscopically

Mixed tumor of the parotid gland contains:
Epithelial cells forming ducts,
Myoepithelial cells,
Chondromyxoid stroma



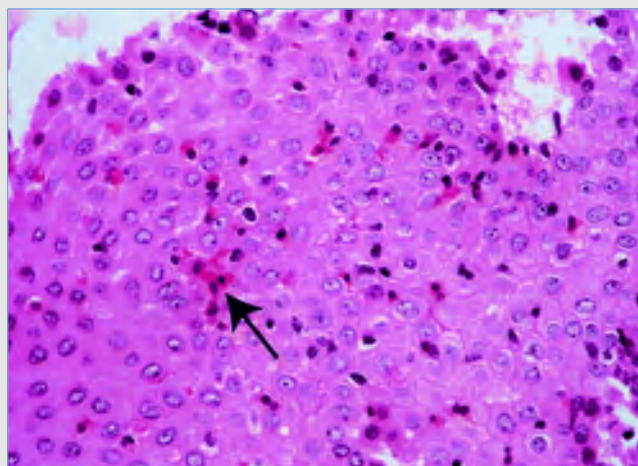
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GASTROESOPHAGEAL REFLUX DISEASE (GERD)



Reflux esophagitis:

necrosis of esophageal epithelium causing ulcers near the junction of the stomach and esophagus



Intraepithelial eosinophils (arrow)

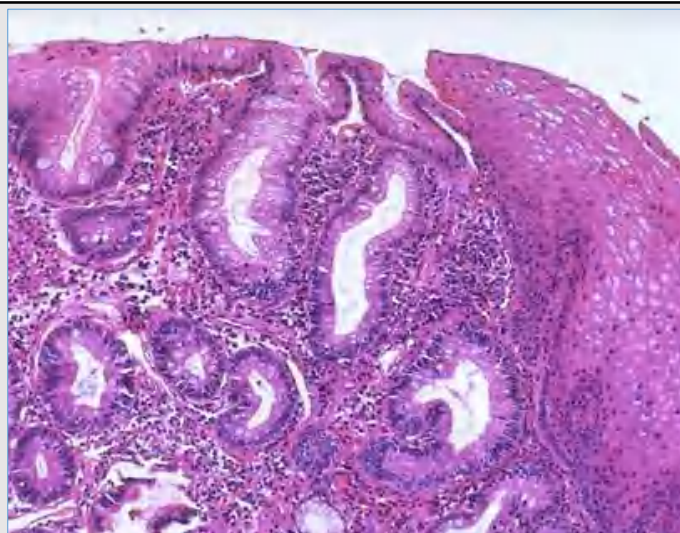
Basal cell hyperplasia (high power, H/E stain).

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BARRETT'S ESOPHAGUS



Areas of mucosal erythema of the lower esophagus, with islands of normal pale esophageal squamous mucosa.

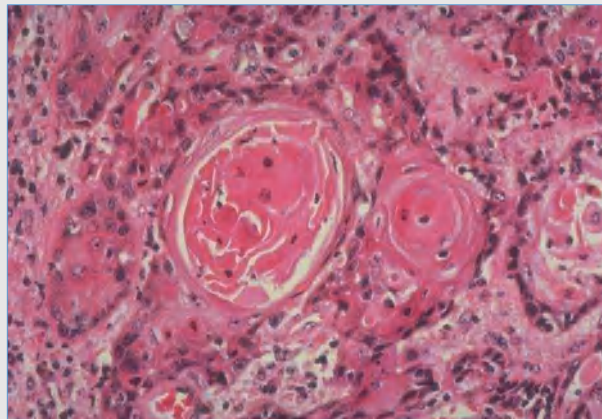


There is gastric-type mucosa above the gastroesophageal junction. Note the columnar epithelium to the left and the squamous epithelium at the right.

Typical Barrett's mucosa shows intestinal metaplasia with chronic inflammation (note the goblet cells in the columnar mucosa).

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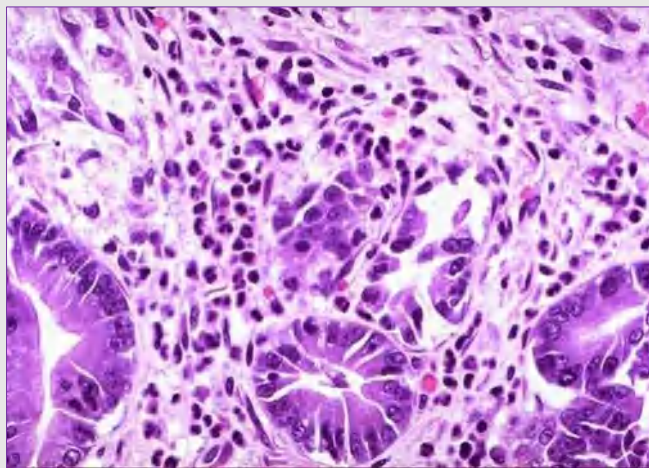
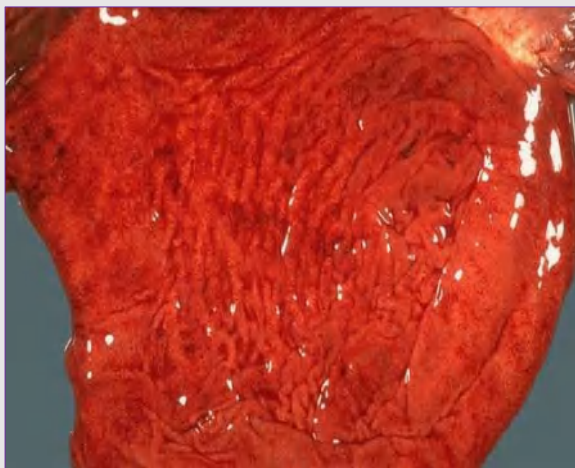
Squamous cell Carcinoma of the Esophagus



- ❖ Whitish thickening of the esophageal wall.
- ❖ The oval structure adjacent to the esophagus represents metastatic squamous cell carcinoma within a lymph node.
- ❖ Infiltrating nests of neoplastic cells.
- ❖ Neoplastic cells having abundant pink cytoplasm and **distinct** cell borders.
- ❖ Keratin pearls (laminated layers).

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Acute Gastritis



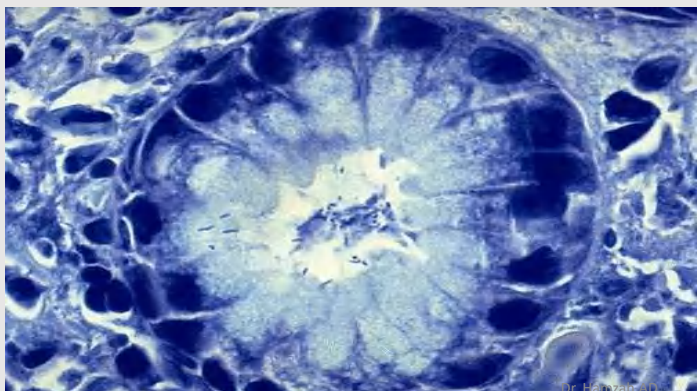
Acute gastritis with a diffusely hyperemic mucosa. There are many causes for acute gastritis: alcoholism, drugs, infections, etc.

Gastric mucosa demonstrates infiltration by neutrophils.

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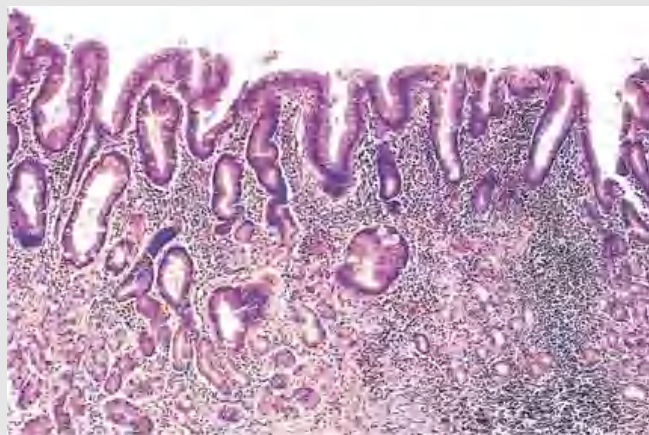
Chronic Gastritis – H Pylori

- ❖ Gastritis is often accompanied by infection with *H. pylori*.
- ❖ Small curved to spiral rod-shaped bacterium is found in the **surface epithelial mucosa of active disease**.
- ❖ The rods are seen here with a methylene blue stain (left) and Silver stain (right).



Chronic Gastritis – H Pylori

- ❖ No erosions, no hemorrhage.
- ❖ Some **neutrophils** (intraepithelial).
- ❖ **Lymphocytes** in lamina propria.
- ❖ Lymphoid follicles hyperplasia in **submucosa**.
- ❖ Coma-shaped eosinophilic figures (*H. Pylori*).
- ❖ Atrophy : mucosal hypoplasia, “thinning”.



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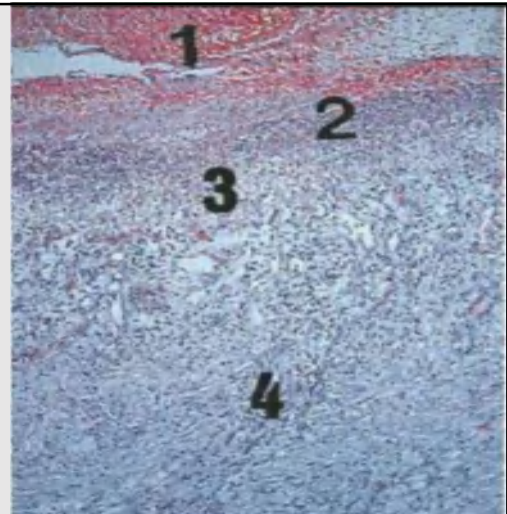
Benign Peptic Ulcer

- “PEPTIC” implies acid causes aggravation.
- Chronic, solitary (usually), adults.
- 80% caused by H. Pylori, NSAIDs, Stress.



- 1- Area of necrosis. 2- Normal thickness.
3- clear base. 4- inverted edges.
5- Radiating spokes.

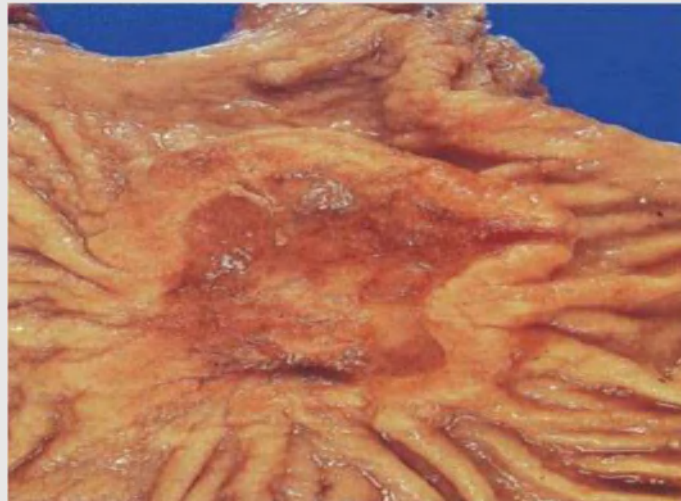
Dr. Hamzah AD



Four zones:

- 1- Necrotic Zone
- 2- area of Acute inflammation
- 3- Granulation tissue
- 4- Dense fibrocollagenous zone.

Malignant Ulcer



Everted and thickened Edges
surrounding the ulcer

Dr. Hamzah AD

Gastric Adenocarcinoma



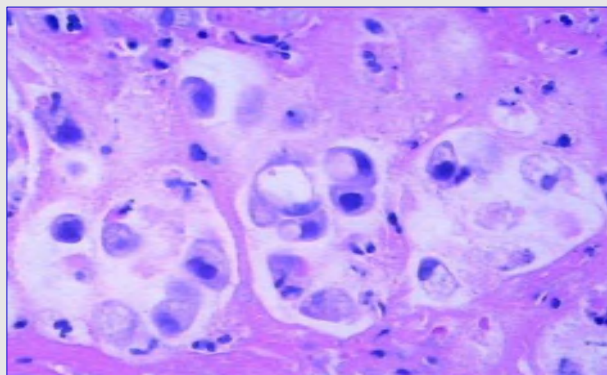
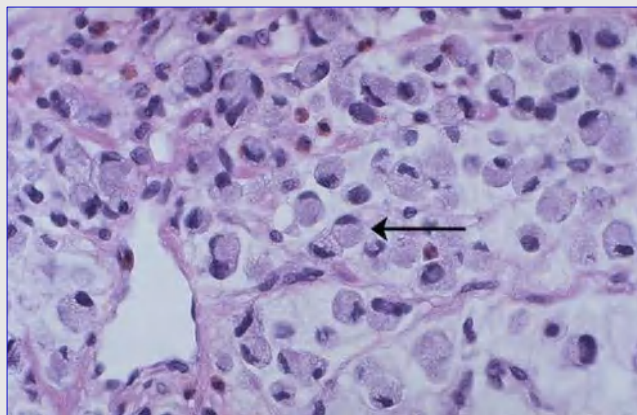
Whitish thickening of the stomach wall

Gastric Adenocarcinoma - Linitis Plastica



- ❖ Linitis Plastica, a diffuse infiltrative gastric adenocarcinoma,
- ❖ A shrunken "leather bottle" appearance.
- ❖ Extensive mucosal erosion.
- ❖ Markedly thickened gastric wall.
- ❖ This has a very poor prognosis.

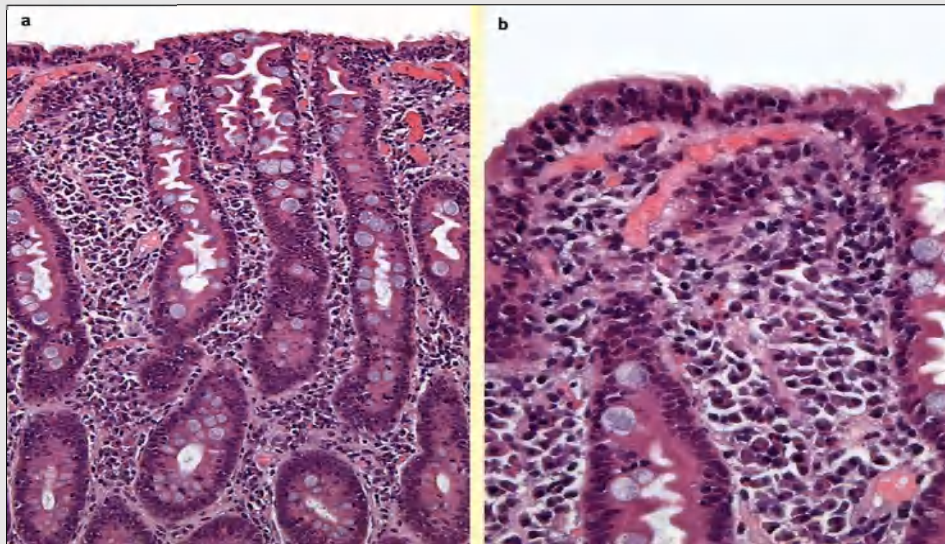
Gastric Adenocarcinoma – Signet ring cell



Poorly differentiated adenocarcinoma, and are often seen **with Linitis Plastica**.

This is a **signet ring cell pattern** of adenocarcinoma in which the cells are filled with **mucin vacuoles** that push the **nucleus to the periphery** giving the cell signet ring appearance.

Celiac Disease (Sprue)



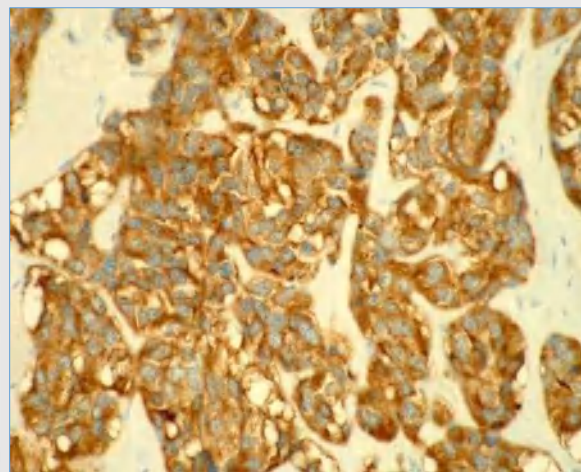
fully developed sprue-type changes. Note the elongated crypts with complete lack (**flattening**) of villi.

Damaged surface epithelium. **intraepithelial lymphocytes.**

Carcinoid tumor of small intestine

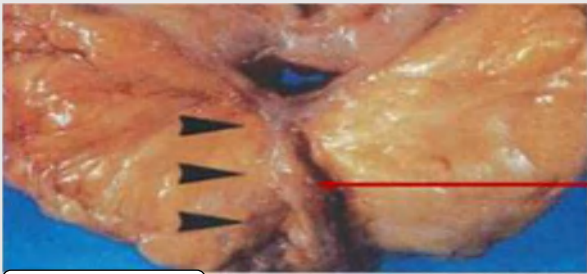


a tumor that has a **faint yellowish color**.
Most are incidental submucosal lesions, though rarely they can be large enough to obstruct the lumen.



Carcinoid tumor showing **strong positive staining** with the **Synaptophysin Immunohistochemical stain (IHC stain)**.
This finding confirms the neuroendocrine nature of this neoplasm.

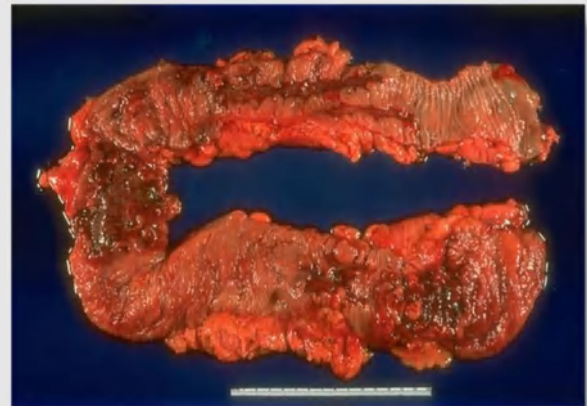
Crohn's Disease



Fissuring

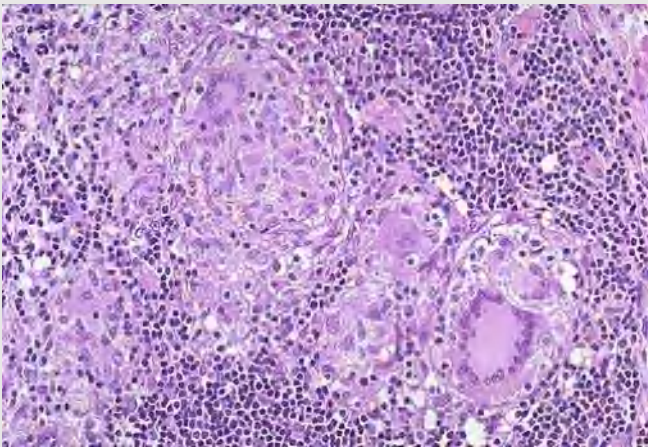


Cobblestone Appearance

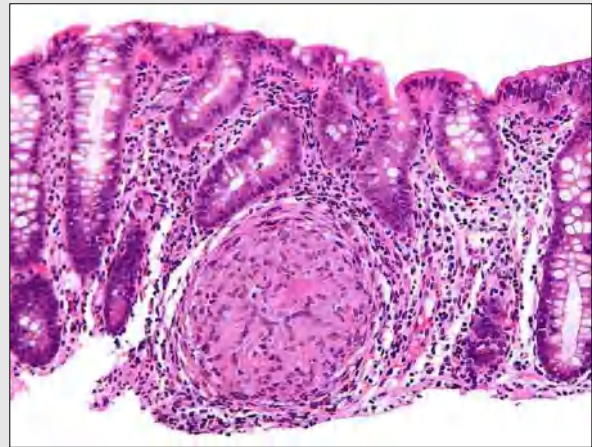


Here the inflammation has produced large, irregularly shaped ulcers that are separated from each other by mucosa that appears close to normal. **(skip Lesions)**

Crohn's Disease

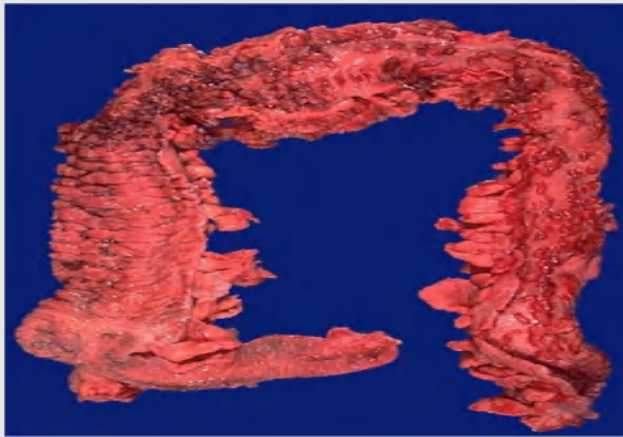


The **granulomatous** nature of the inflammation of Crohn's disease is demonstrated here with **epithelioid** cells, **giant** cells, and **lymphocytes**.
Special stains for organisms are **negative**.



All layers of intestinal wall show transmuritis with chronic inflammatory cell infiltrate, lymphoid aggregates and mild fibrosis. Few epithelioid granuloma are seen.

Ulcerative Colitis

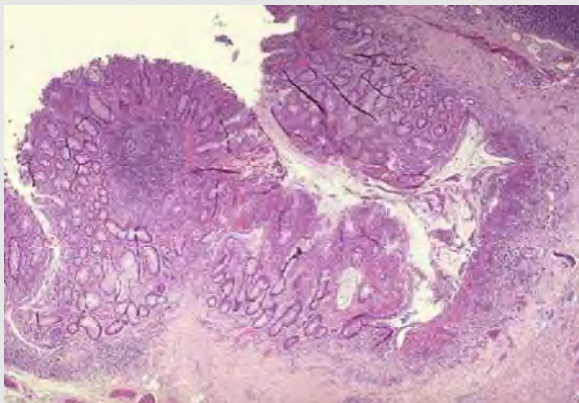


The most intense inflammation **begins** at the **sigmoid** colon (Right) and extends **upward** and around to the ascending colon.
At the lower left is the ileocecal valve with a portion of terminal ileum that is **not involved**.

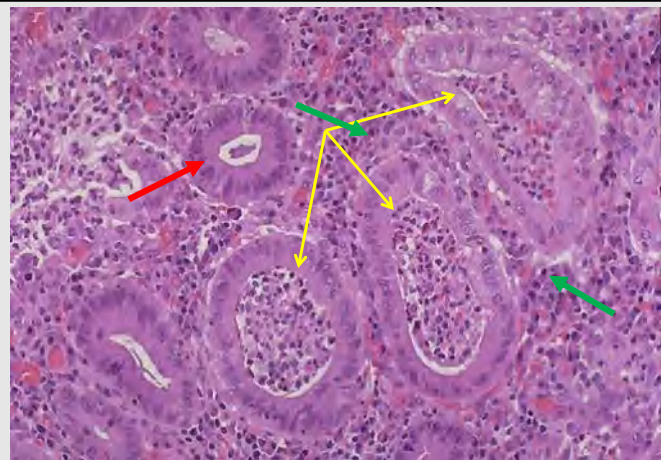


Pseudo-polyps are seen here in a severe case.
The remaining mucosa has been ulcerated away and is hyperemic.

Ulcerative Colitis



The inflammation is **confined to the mucosa**.
Here, the mucosa is eroded by an ulcer that undermines surrounding mucosa.



Crypt abscesses (neutrophilic exudate is found in glandular lumens) are typical histologic finding.
The submucosa: intense inflammation with **plasma cells**.
The glands: **depletion of goblet cells**.

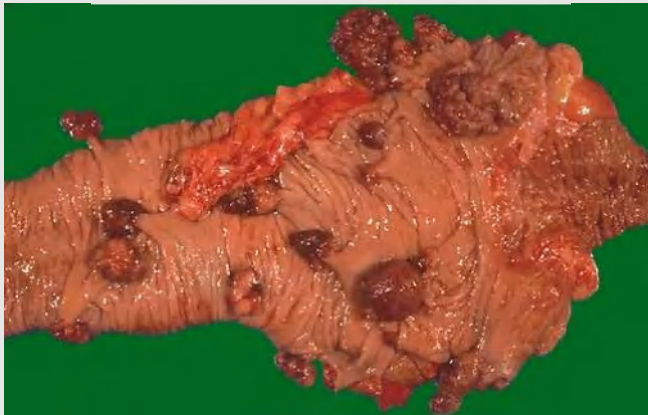
Ulcerative Colitis with Dysplasia



Over time, there is a risk for adenocarcinoma with ulcerative colitis.

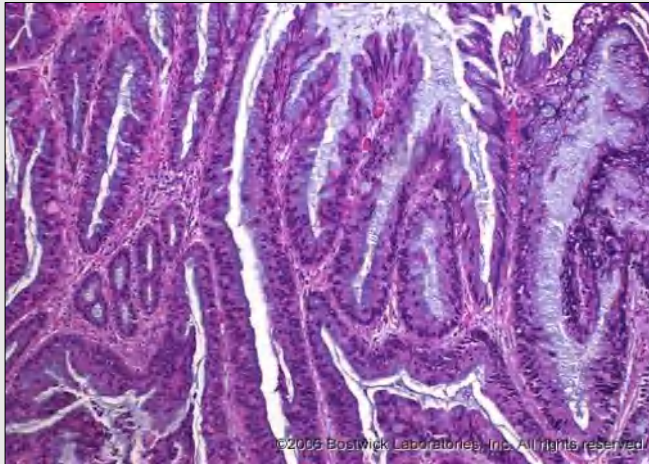
Here, **normal** glands are seen at the **left**, but glands at the **right** have **dysplasia**.

Adenomatous polyp of the colon



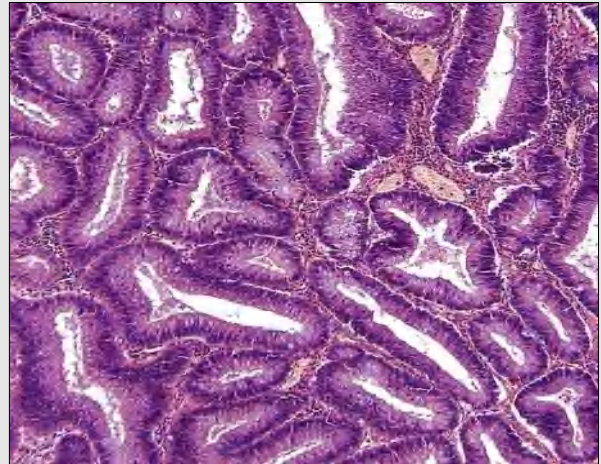
Multiple adenomatous polyps of the cecum are seen here in a case of **familial adenomatous polyposis**, a genetic syndrome in which an abnormal genetic mutation leads to development of multiple neoplasms in the colon.

Adenomatous polyp of the colon



Villous adenomas

more aggressive than tubular adenomas. They have a HIGHER rate of developing into adenocarcinomas than “tubular”.



TUBULAR adenoma

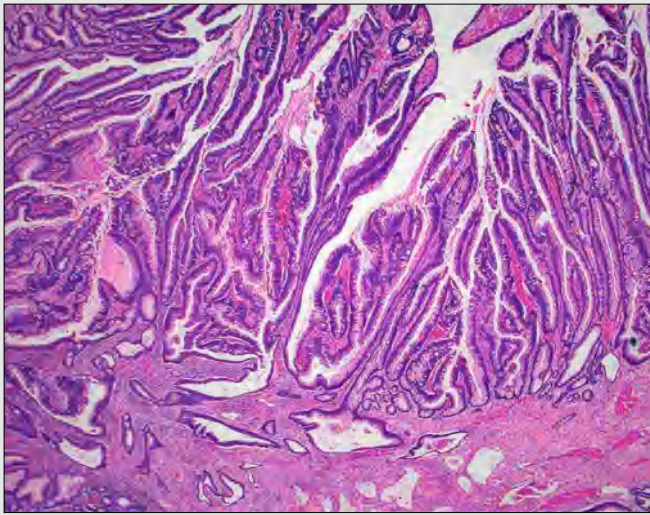
crowded dysplastic glands and chronic inflammation.

Adenocarcinoma of the Colon

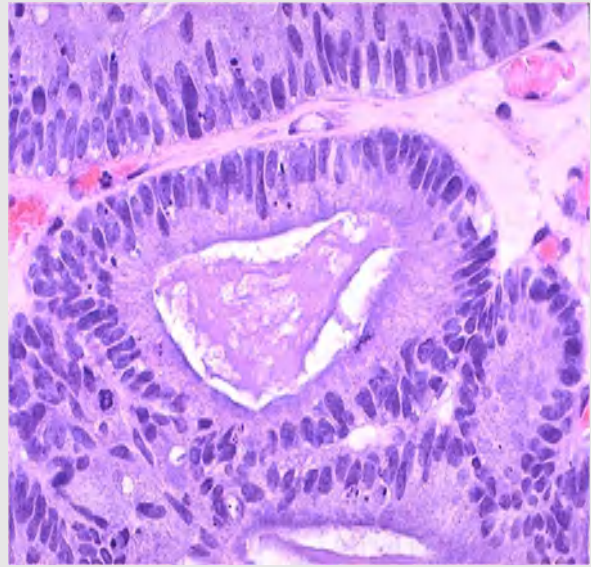


This is an adenocarcinoma arising in a villous adenoma.
The surface of the neoplasm is **polypoid** and **reddish** pink.
Hemorrhage from the surface of the tumor creates a positive stool.

Adenocarcinoma of the Colon



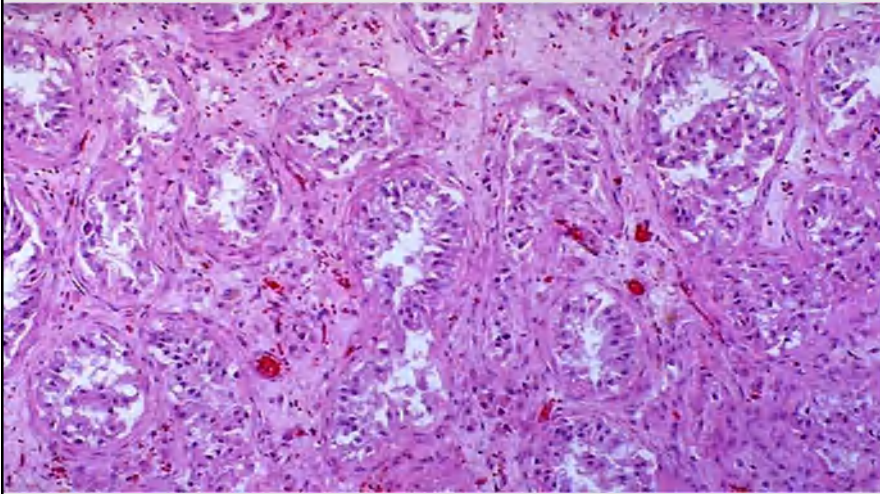
Tumor consists of crowded irregular malignant acini separated by thin fibro-vascular stroma.



The neoplastic glands of adenocarcinoma have **crowded nuclei with hyperchromatism and pleomorphism. No normal goblet cells** are seen

Male Genital system

Testicular Atrophy



Marked loss of germ cells with remaining tall pink Sertoli cells, peritubular fibrosis, and interstitial fibrosis. If generalized, this is a cause for infertility.

Testicular Torsion



This testis has undergone infarction following testicular torsion.

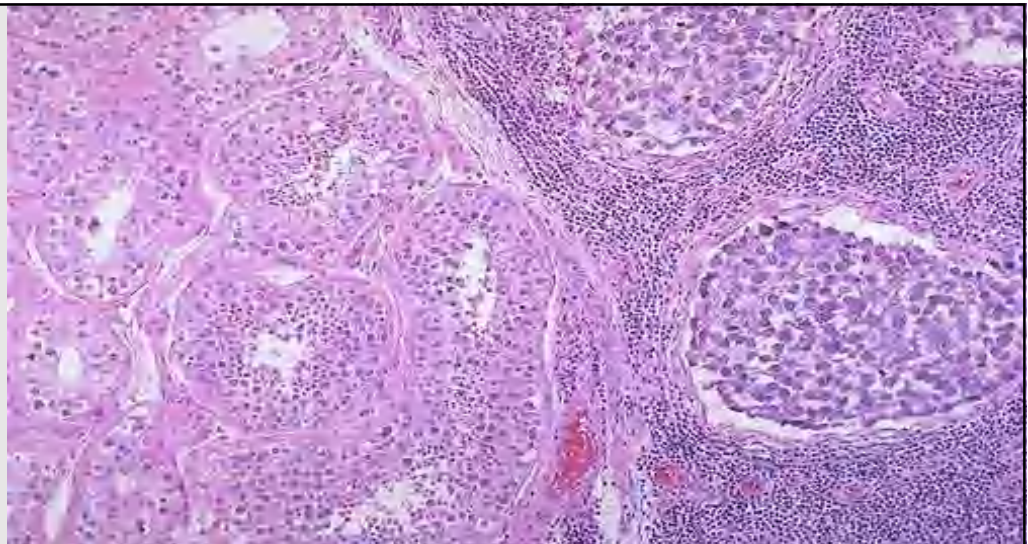
Torsion is an uncommon condition, but a medical emergency. It occurs when twisting of the spermatic cord cuts off the venous drainage, leading to hemorrhagic infarction.

Seminomas (Germ cell Tumor)



Large, Soft. Well demarcated, Gray-white tumour. Usually homogeneous.

Seminomas



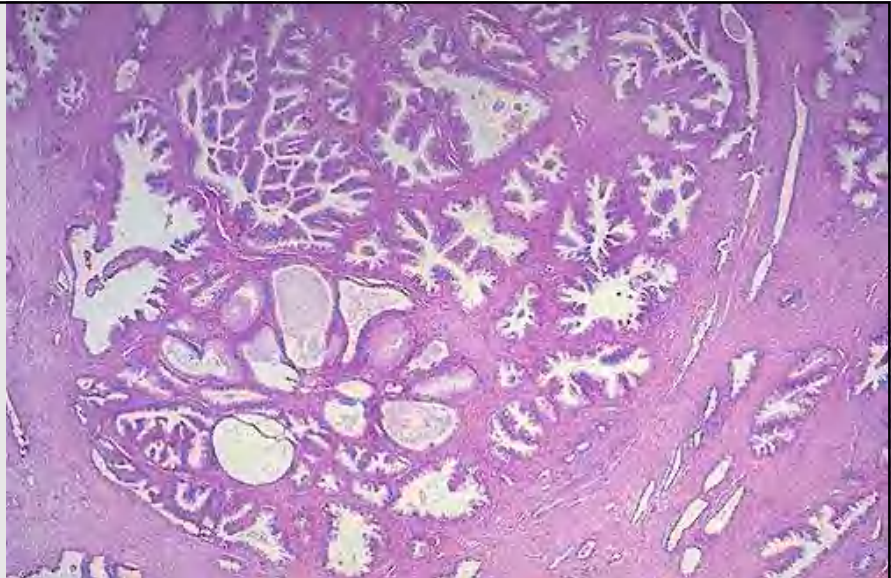
Large cells with Distinct cell borders. Clear (glycogen rich) cytoplasm, Rounded nuclei with nucleoli. The cells are often arranged in small lobule, separated by fibrous septa. Lymphocytic infiltrated is usually present and may form granuloma

Benign prostatic hyperplasia



Enlarged irregular prostate. The cut surface contain multiple well circumscribed nodules. The nodule can be solid or cystic.

Benign prostatic hyperplasia



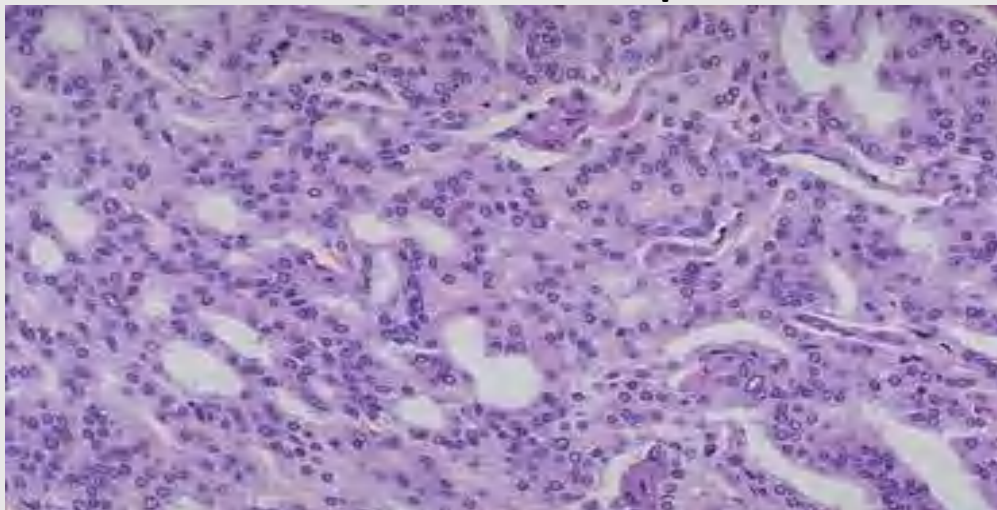
Proliferation of the gland and fibromuscular stroma.
Crowding of the epithelial cells → papillary projections.
The lumen shows corpora amylacea.

Adenocarcinoma of the prostate



Cut section : firm, gray white to yellow., infiltrating adjacent gland, ill-defined margin.

Adenocarcinoma of the prostate



Well-differentiated carcinoma:

Small glands infiltrating the stroma. Back to back glands not encircled by stroma.
The neoplastic gland are single layer of cuboidal cells with absence of the basal cells.

Female Genital System

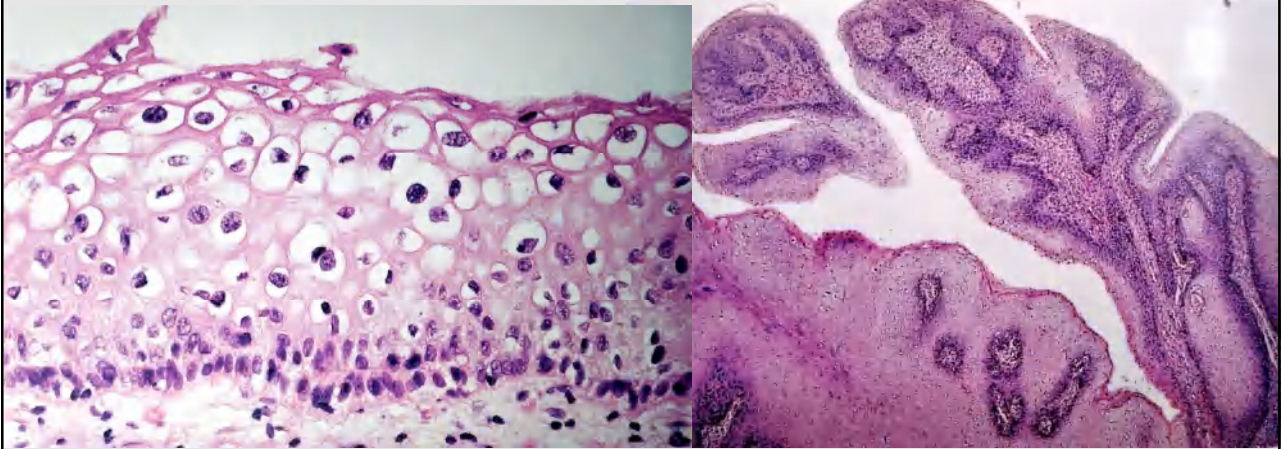
Condyloma accuminatum

Due to human papillomavirus (HPV 6 - 11).
Benign but may be precancerous.

Gross
wart

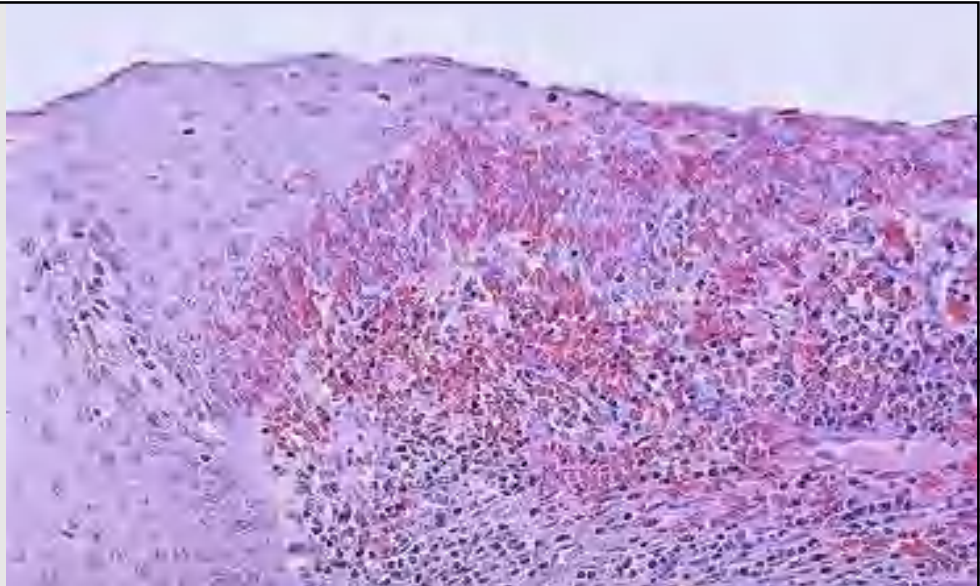


Condyloma accuminatum



- Papillomatous proliferating squamous epithelium.
- Koilocytosis (halo cells):
Nuclear enlargement, irregular nuclear membrane, hyperchromatosis, perinuclear halo.

Cervicitis

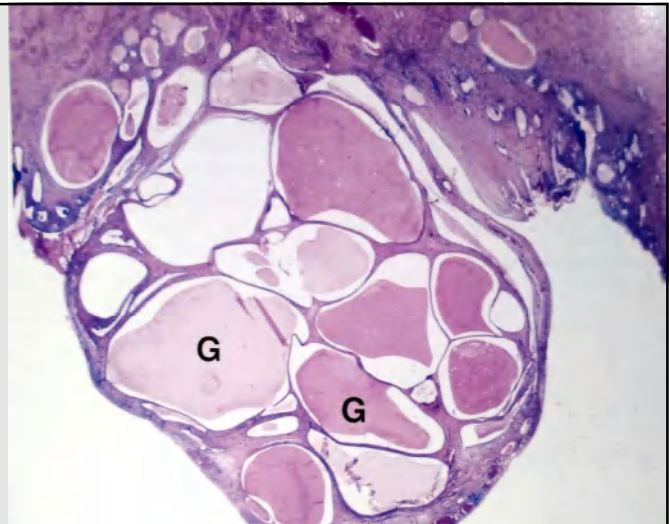


Chronic cervicitis Chronic inflammatory cells infiltration, Nabothian cysts (mucus-filled cyst).
Endocervix squamous metaplasia.

Endocervical polyp



Polypoid protruding mass, Soft, Covered by smooth glistening surface.



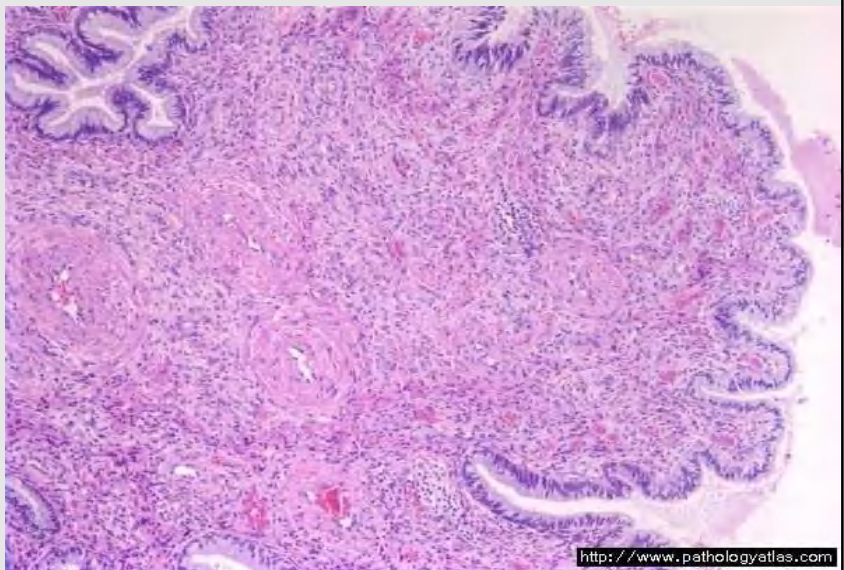
Microscopic:

- Dilated endocervical glands.
- Congested stroma
- Inflammatory cells +/-

Endocervical polyp

Microscopic:

- Dilated endocervical glands.
- Congested stroma
- Inflammatory cells +/-



Adenomyosis

Microscopic

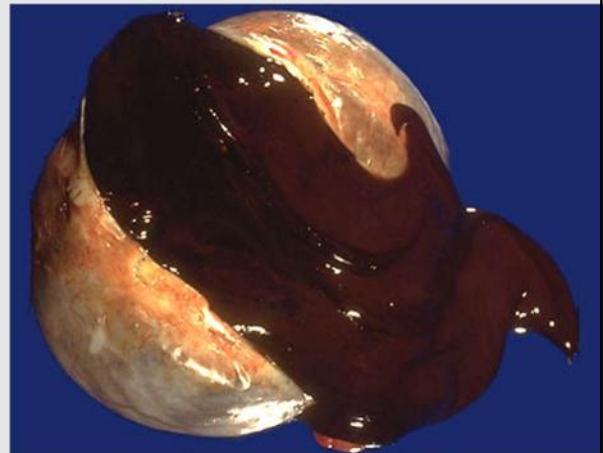
- Endometrial glands & stroma within myometrium.
- Blood within glands +/-
- Hemosiderin-laden macrophages (chronic congestion and hemorrhage).



Endometriosis (Endometriosis externa)

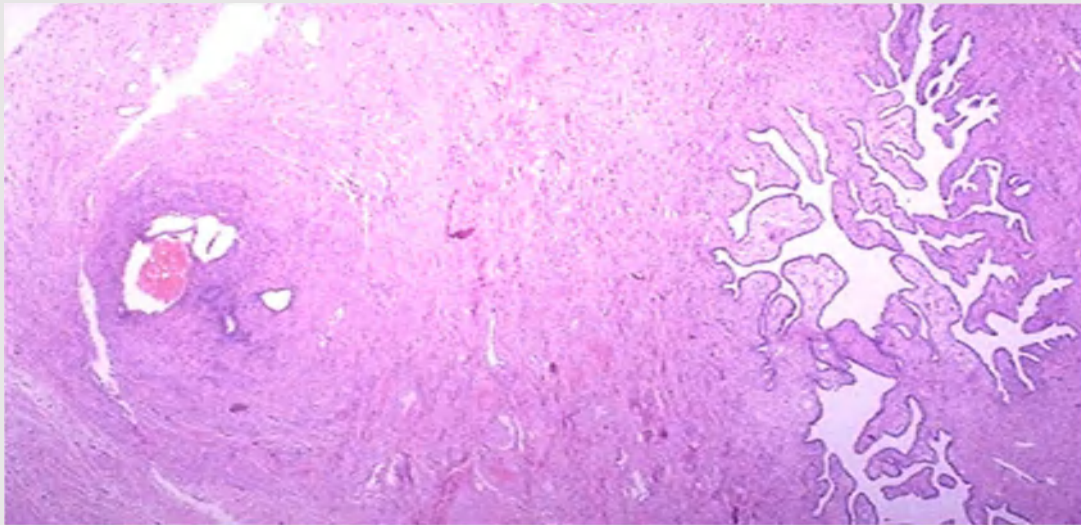


Red-blue to yellow brown nodules due to blood collection in these foci.



In the ovary: the lesion is large, blood filled cyst called **chocolate cysts** (old blood).

Endometriosis (Endometriosis externa)



a small cluster of endometrial glands and stroma with hemorrhage are seen at the left near the surface of the fallopian tube. The lumen of the tube is at the right.

Endometriosis

- ▶ Endometrial glands (**G**)
- ▶ Endometrial stroma (**S**)
- ▶ Hemosiderin-laden macrophages (**M**)



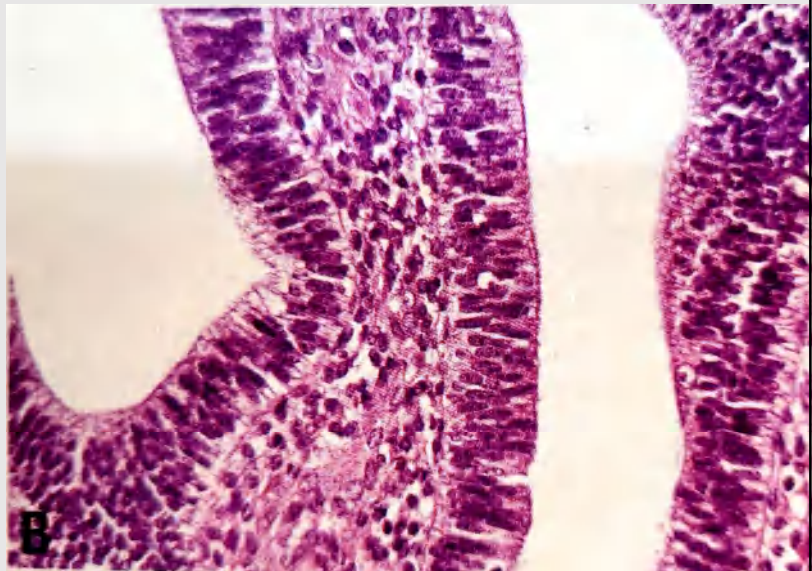
Simple cystic endometrium hyperplasia

- Irregular dilated glands (G)
- Variation of gland size
- Cysts (C)
- Atypia (-)
- Simple end. hyperplasia without atypia



Simple cystic endometrium hyperplasia

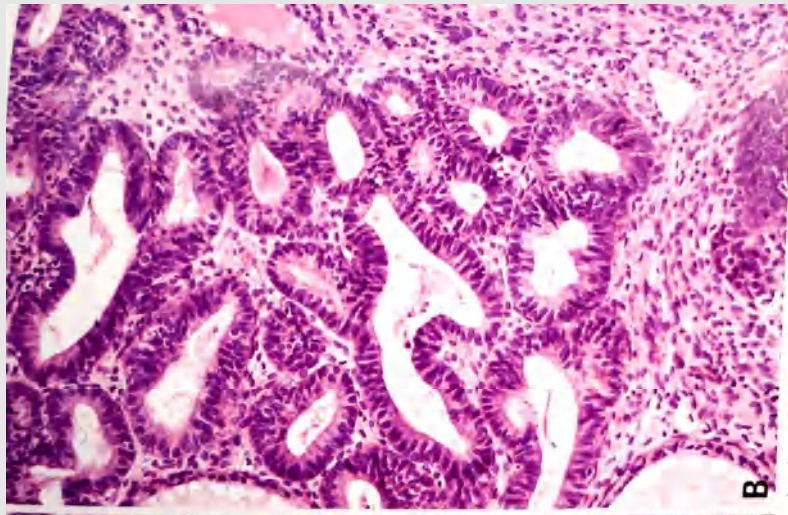
- Irregular dilated glands (G)
- Variation of gland size
- Cysts (C)
- Atypia (+)
- Simple end. hyperplasia with atypia



Complex endometrium hyperplasia

Microscopic:

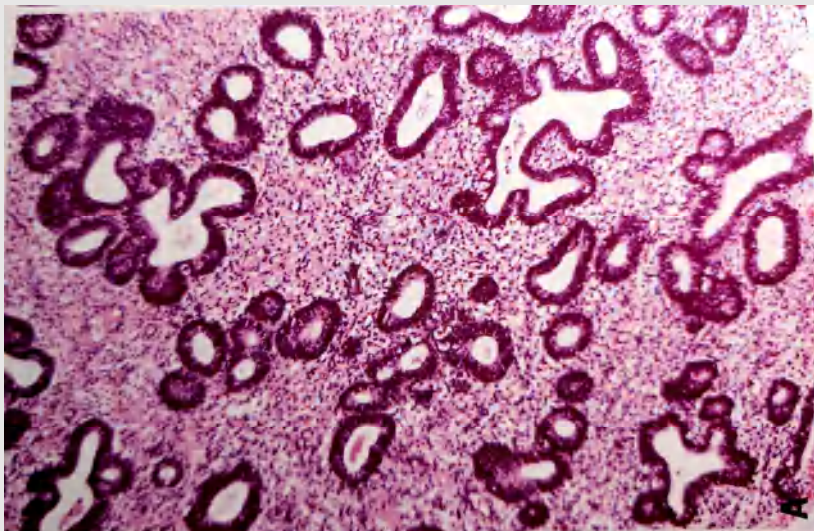
- Increase in size & endometrium glands
- **Irregular glands with Budding**
- Atypia (-)
- complex end. hyperplasia without atypia



Complex endometrium hyperplasia

Microscopic:

- Increase in size & endometrium glands
- **Irregular glands with Budding**
- Atypia (+)
- complex end. hyperplasia with atypia

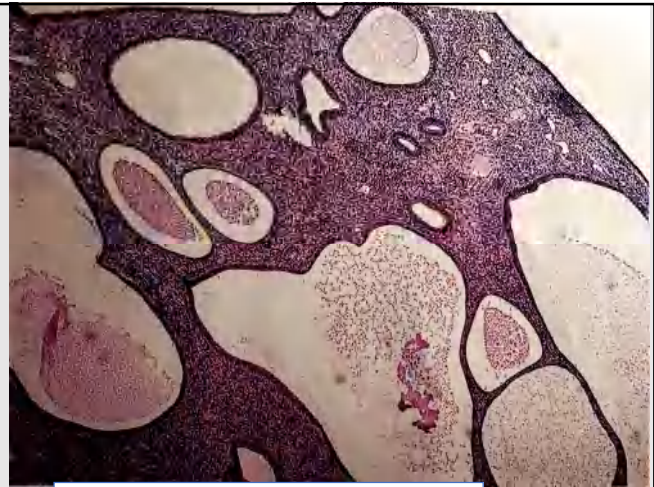


Endometrium polyp



Gross:

- Polypoid mass in the endometrial cavity
- Measured up to 10 cm.



Microscopic

- Large blood vessels
- Fibrotic stroma
- Proliferative endometrial glands
- Cystic dilated glands

Leiomyoma

Gross

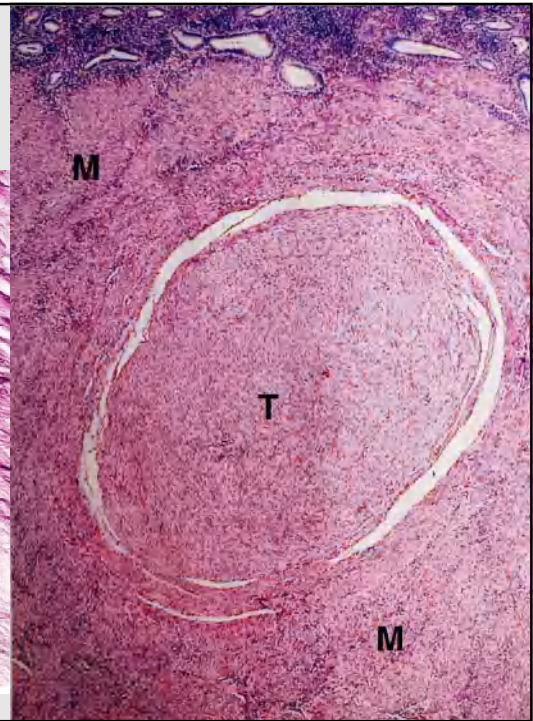
- Asymmetric uterus enlargement.
- Solitary or multiple.
- Site: submucous, interstitial and subserous.
- **C/S:** white whirly appearance



Leiomyoma

Microscopic >>>

- Well defined nodule



Leiomyosarcoma

Gross:

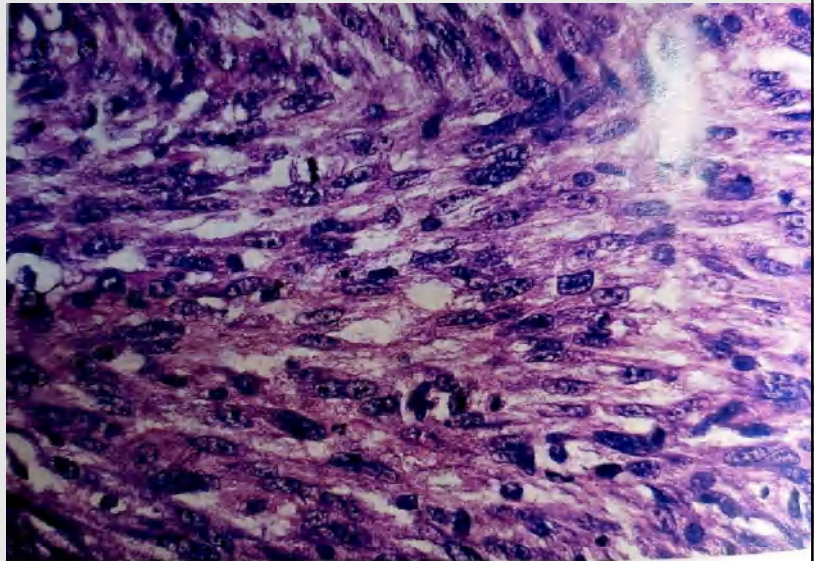
- Large uterus
- Polypoid mass
- Necrosis
- hemorrhage



Leiomyosarcoma

Microscopic:

1. Ugly cytological features
2. Atypical mitosis



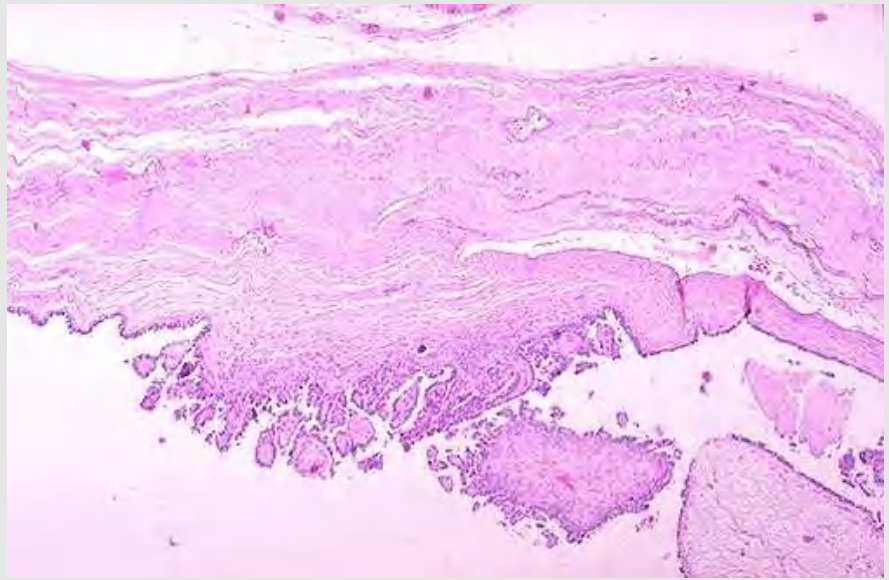
Serous tumors

Bilateral & benign.

Usually small (5-10 cm)
sometimes large (up to 40 cm) in diameter,
cystic (so called cystadenoma,
cystadenocarcinoma),
Filled with serous fluid.
The lining epithelium is smooth.



Serous tumors

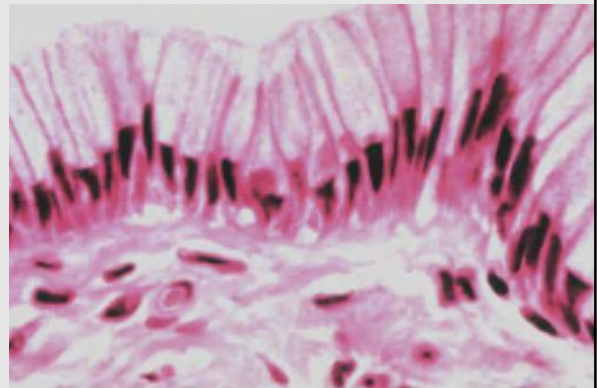


Benign tumors: Lined by single layer of columnar layer partially ciliated and partially secretory, papillated, psamoma bodies may be seen in the papillae.

Mucinous tumors

Gross: **Unilateral**, huge size, Usually cystic (rarely solid), Filled with bluish mucinous material

Microscopic: The lining epithelium is single layer of tall columnar epithelium that shows apical vacuolation and absence of cilia, with rare papillae.



Benign (mature) cystic teratoma

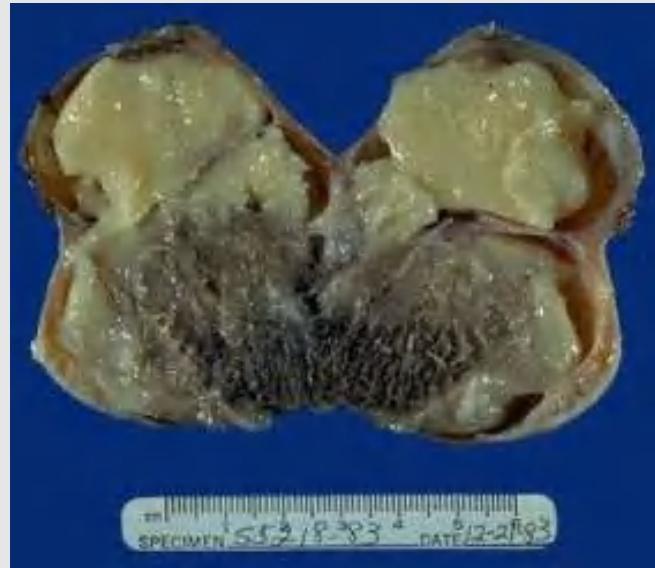
Unilateral 90% of cases.

Cystic.

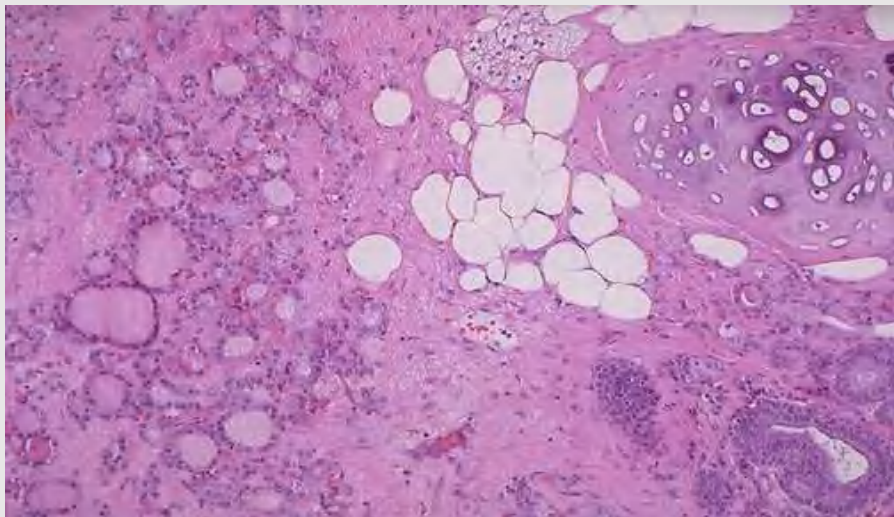
Rarely exceed 10 cm in diameter.

Cut section Filled with sebaceous secretion and may show hair.

Sometimes teeth protrude.

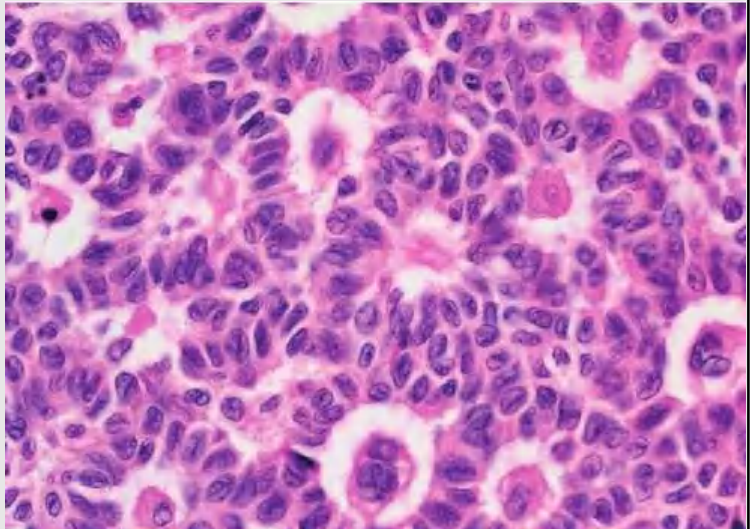


Benign (mature) cystic teratoma



The cyst lined by stratified squamous epithelium with sebaceous gland and hair follicles, Sometimes, bone, cartilage, teeth thyroid ... etc. all of these tissues are mature.

Granulosa stromal tumor



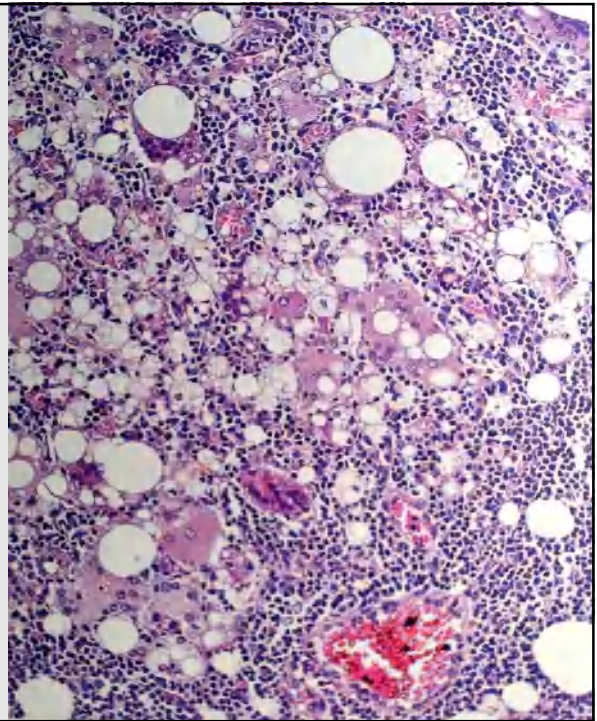
Sheets or cords of tumor cells, Small cuboidal to polygonal cell with coffee bean nuclei, Gland-like structures filled with acidophilic material (Call-Exner bodies).

Breast

Granulomatous Mastitis

Giant cells
Epithelioid cells

What else?

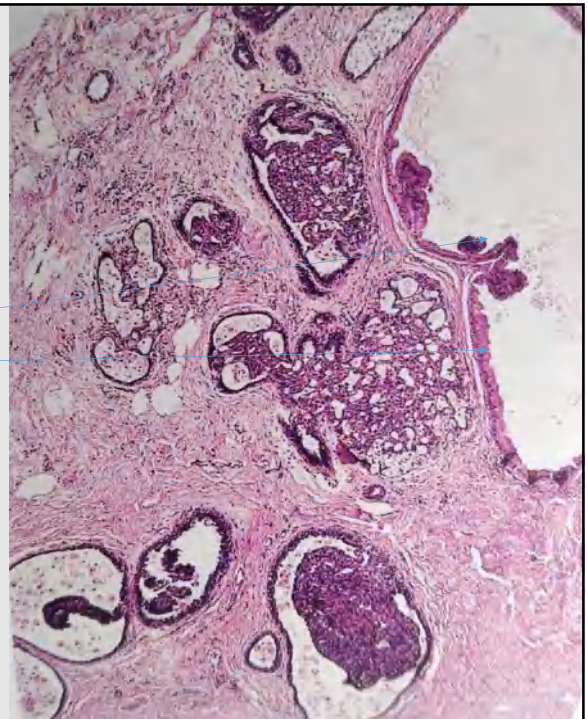


FIBROCYSTIC DISEASES

intracystic papillary structure

Apocrine metaplasia

hyperplasia

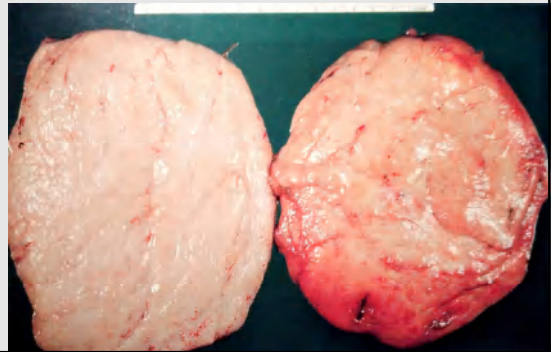
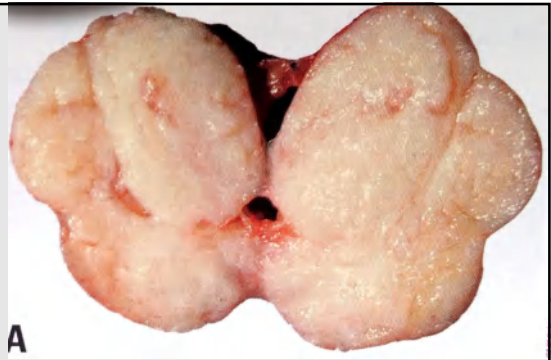


FIBROADENOMA

unilateral
women between the ages of 15 and 35
Convert to malignancy...very rare.

Gross

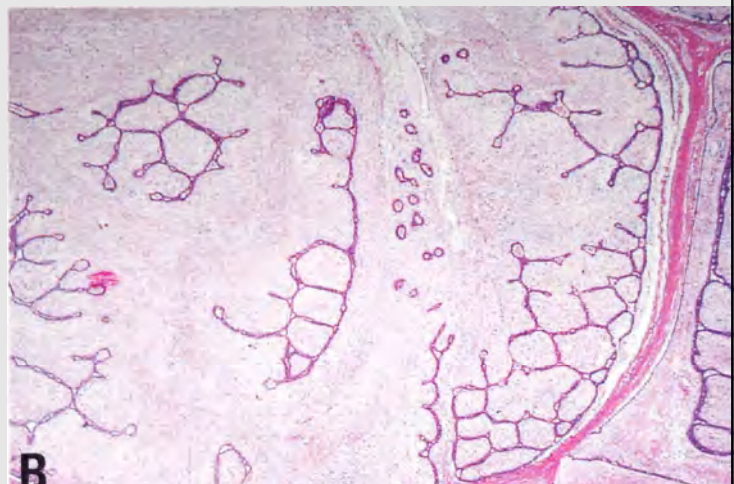
- Firm non attached to skin (**mobile**)
- **Capsulated**
- Small to large size 3-20 cm in diameter.
- **C/S**: white to yellow surface.



FIBROADENOMA

Microscopic

- ▶ **Biphasic (Fibro-epithelial)** tumor **with capsule**.
- ▶ Formed of **ducts** and **stroma**.
- ▶ **Ducts lined by double epithelium.**
- ▶ Stroma shows cellular hyperplasia.
- ▶ Other stroma componentes +/- :
 - **Percanalicular pattern.**
 - **Intracanalicular pattern.**



FIBROADENOMA

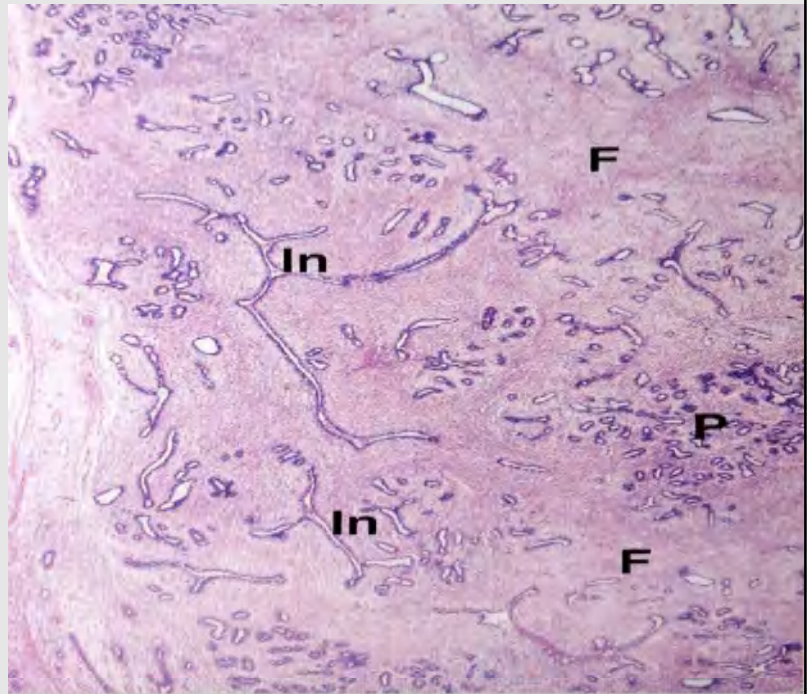
1. Intracanalicular pattern. (In):

The stroma compresses and distorts the glands into cleft-like spaces in the intracanalicular pattern.

2. Pericanalicular pattern. (P):

The stroma surrounds the glands, without distorting them, and the glands maintain their open lumens in the pericanalicular pattern.

3. Fibrous tissue (F)



PHYLLOID TUMOR

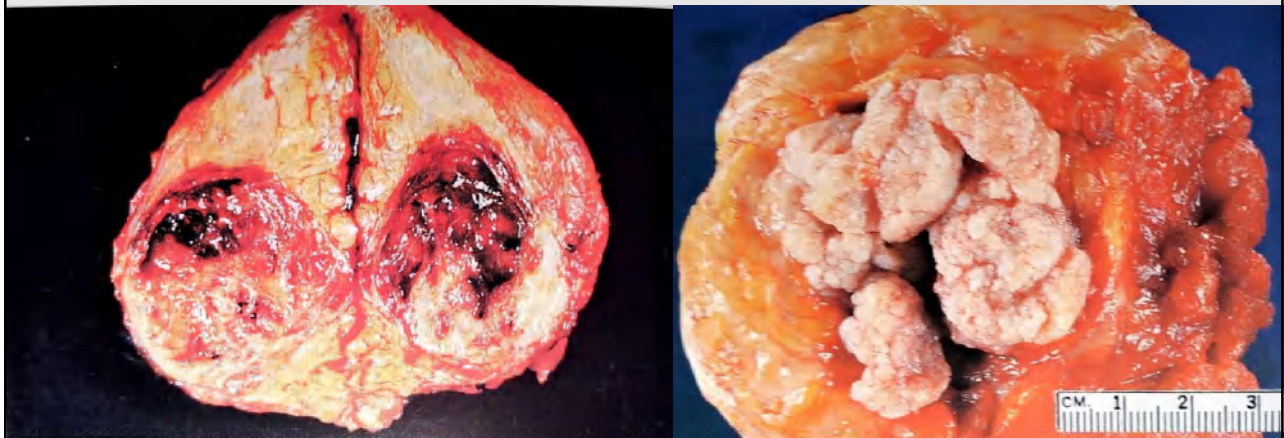
- One of the breast stromal tumor.
- Usually it's unilateral
- Bloody nipple discharge.

Gross

Well formed mass with capsule. Firm.

C/S: shows leaf-like appearance.

White to yellow appearance. Necrosis +/-



PHYLLOID TUMOR

Microscopic

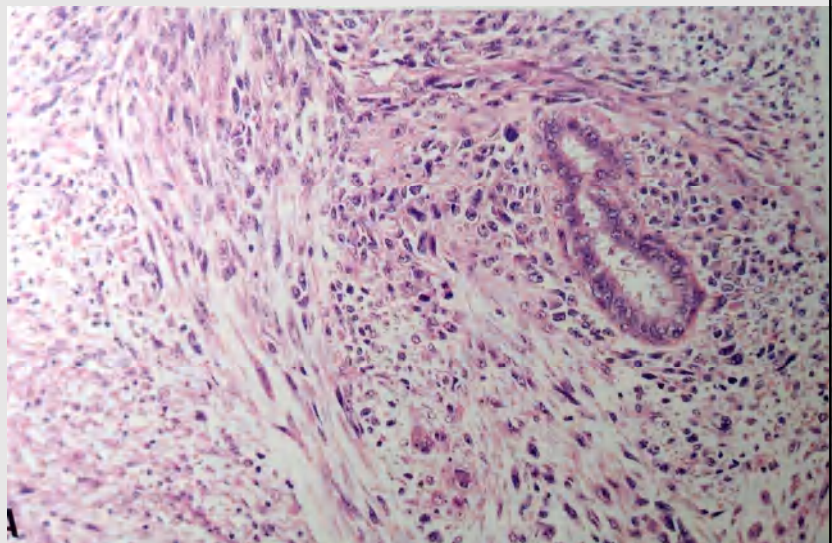
- Biphasic tumor
- Ducts are compressed to **cysts**
- Lined by double layer of cells.
- Invagination of ducts.
- Stroma shows **spindle cells hyperplasia**.
- **Metaplastic stroma +/-**



PHYLLOID TUMOR

-Malignant! (cystosarcoma)

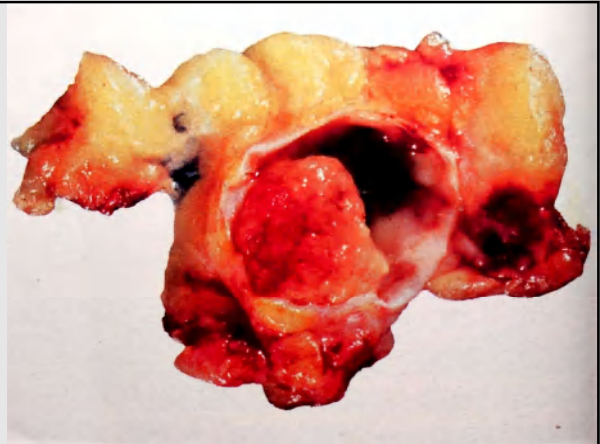
- Spindle cells
- Pleomorphism
- Hyperchromatic nuclei
- mitosis



INTRADUCTAL PAPILLOMA

Gross

- **Small** nodules not exceeded 3cm in diameter.
- **C/S:** shows cyst with intra lumen projecting mass (Cauliflower appearance).



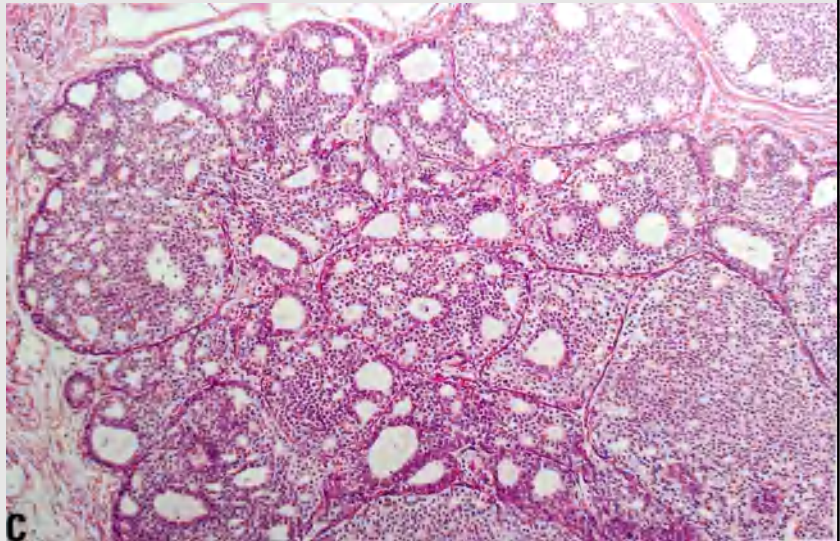
Microscopic

- Intracystic papillary growth (**Arborescent structures**)
- **Lined by double epithelium:**
 - outer Myoepithelial cells
 - inner epithelial cells.

Ductal Carcinoma In Situ (DCIS)

Types

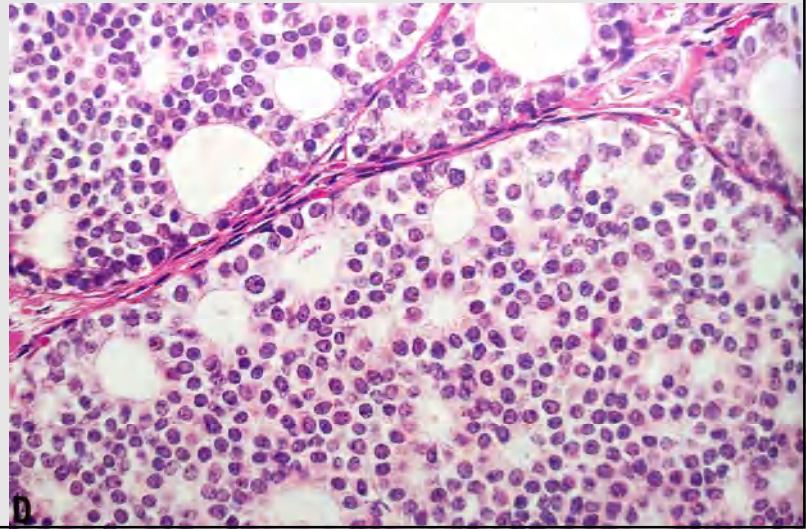
1. **Solid.**
2. Cribriform (**necrosis**)
3. Micropapillary.
4. Flat.
5. Clear cells
6. Apocrine
7. Comedo



Ductal Carcinoma In Situ (DCIS)

Types

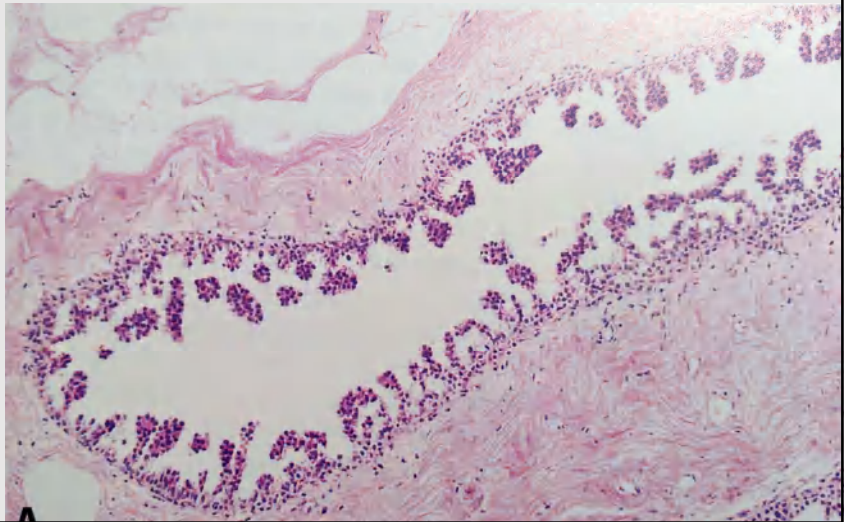
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Ductal Carcinoma In Situ (DCIS)

Types

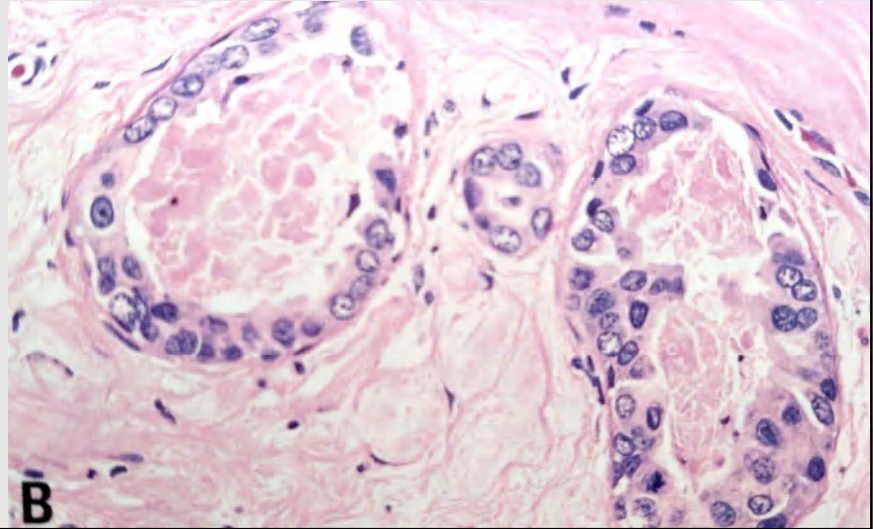
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Ductal Carcinoma In Situ (DCIS)

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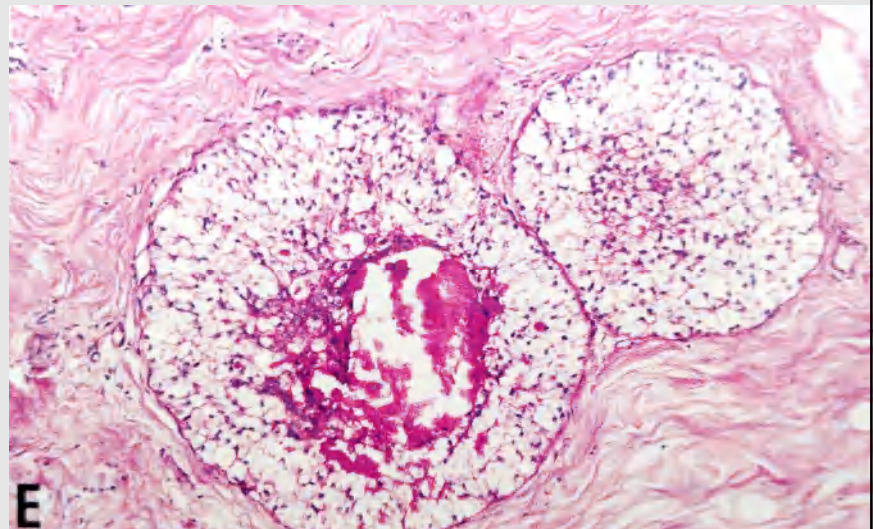
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Ductal Carcinoma In Situ (DCIS)

Types

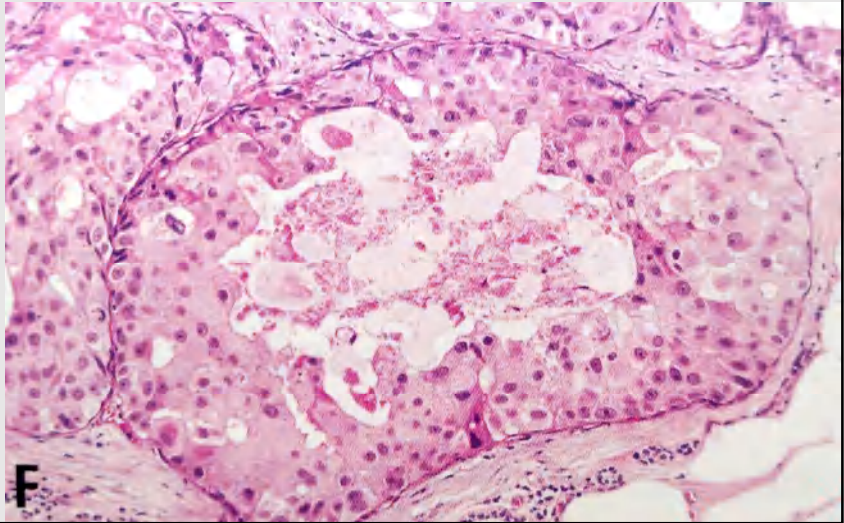
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Ductal Carcinoma In Situ (DCIS)

Types

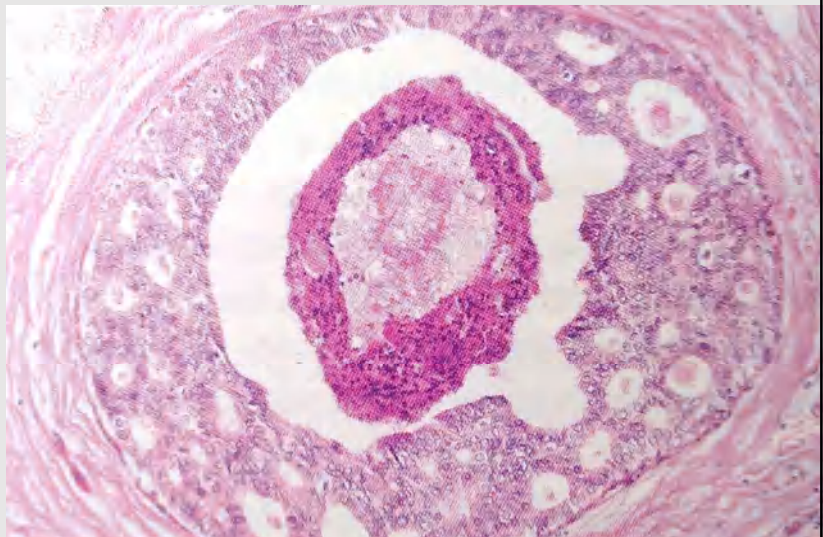
1. Solid.
2. Cribriform
3. Micropapillary.
4. Flat.
5. Clear cells
6. **Apocrine**
7. Comedo



Ductal Carcinoma In Situ (DCIS)

Types

1. Solid.
2. Cribriform
3. Micropapillary.
4. Flat.
5. Clear cells
6. Apocrine
7. **Comedo**



Invasive Ductal Carcinoma(IDC)

- Invading basement membrane into the stroma.
- No Myoepithelial cells.
- **Represent 90% of breast carcinoma.**

Grades:

1. Well differentiated.
2. Moderate differentiated.
3. Poor differentiated.

Depends on score of 3-9/9

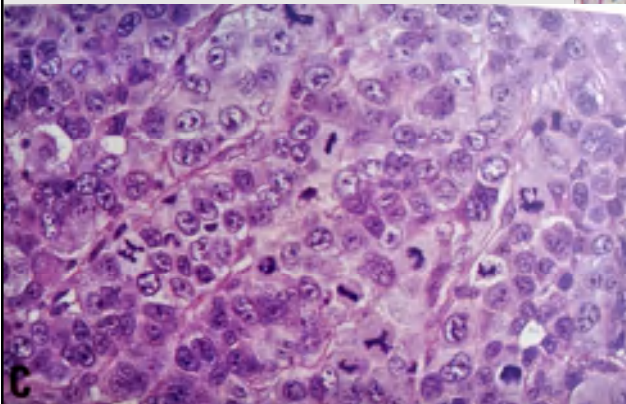
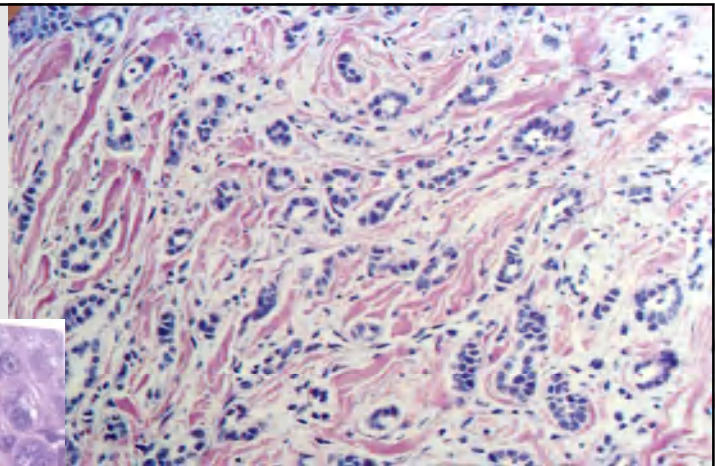
- TUBULE (duct) 3
- NUCLEAR PLEOMORPHISM 3
- MITOSIS 3



Invasive Ductal Carcinoma(IDC)

Microscopic Features >>>>

- Irregular ducts infiltrating stroma
- One layer atypical epithelial cells

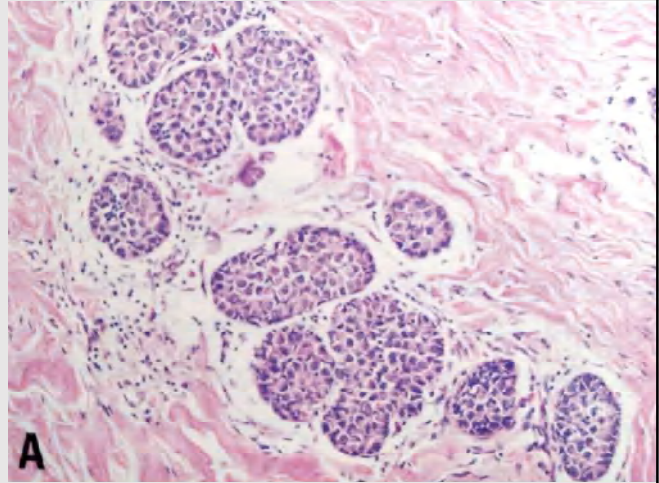


<<<< Microscopic Features

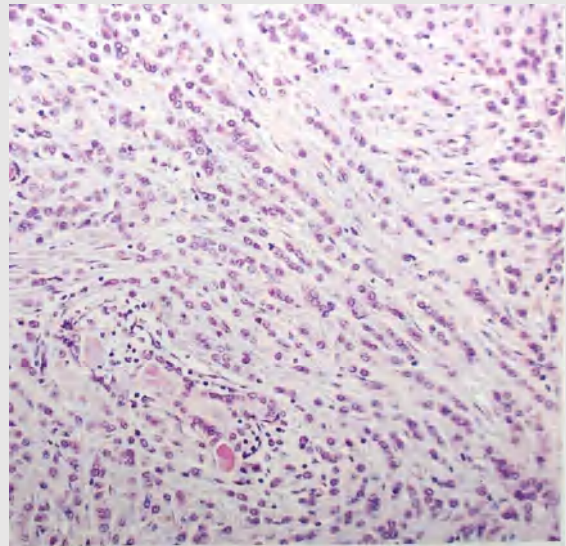
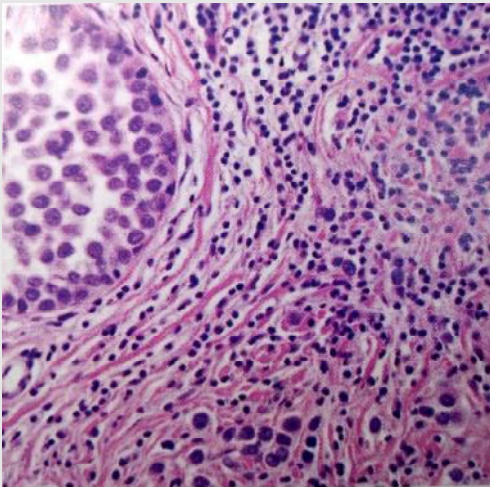
- Cellular Pleomorphism
- Mitosis
- Hyperchromatic nuclei

Lobular Carcinoma In Situ (LCIS)

- It's malignant growth within terminal ductal Lobular unit.
- Not invading basement membrane into the stroma.
- **Monomorphic round tumor cells** with **indistinct cell membrane**.
- No necrosis or calcification.

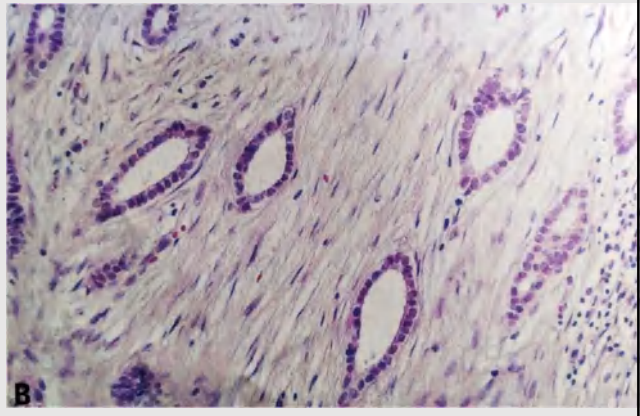
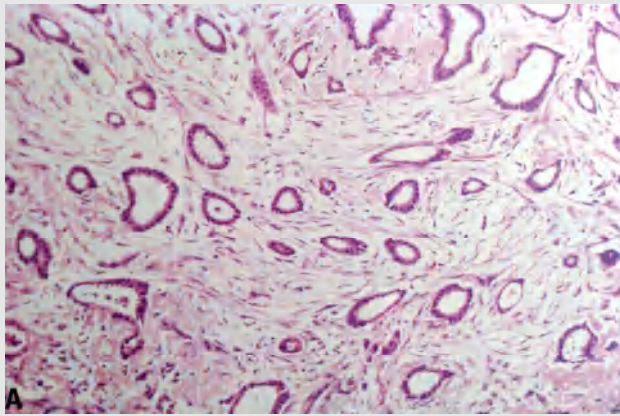


Invasive Lobular Carcinoma (ILC)



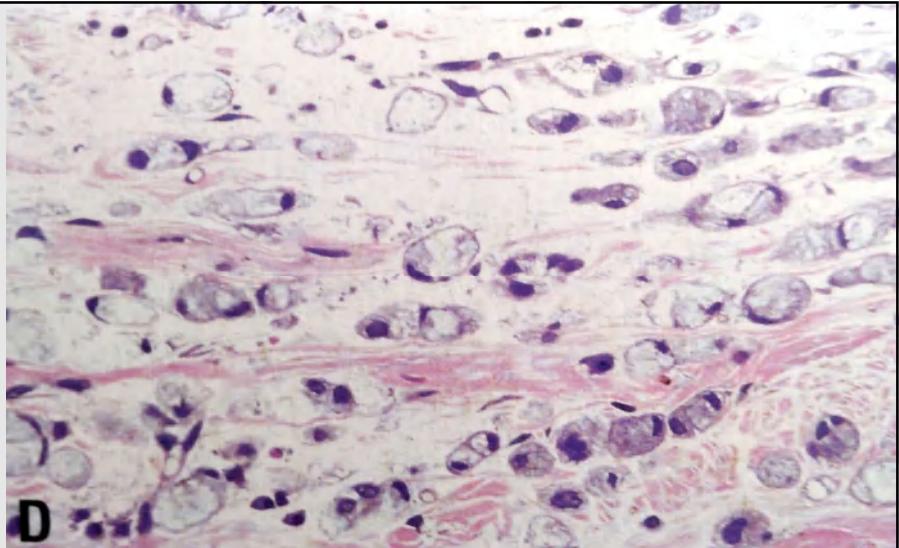
- Associated usually with LCIS.
- Formed of small tumor cells arranged singly in cord-like (**Indian file pattern**) **Spot diagnosis**
- Tumor cells show minimal atypia.

Tubular carcinoma



- Formed of **angulated** ducts.
- Lined by **single** layer of epithelium.
- Not associated with necrosis or mitosis.

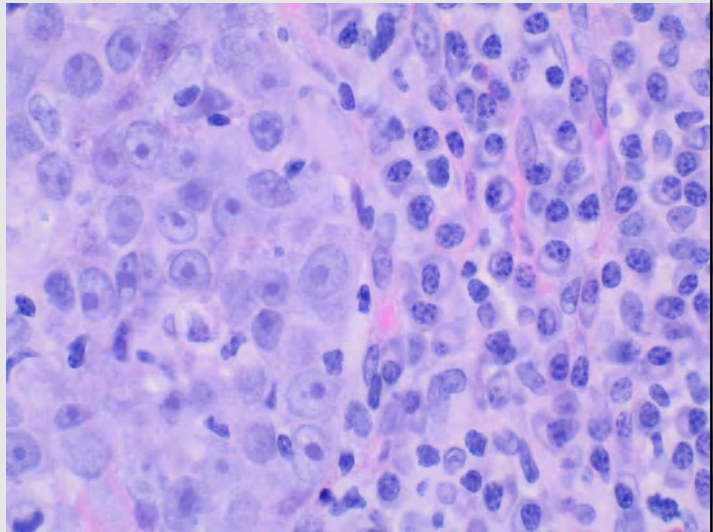
Mucinous carcinoma



- Microscopic:
Clusters of tumor cells floating in mucin pool.
Signet Ring Appearance

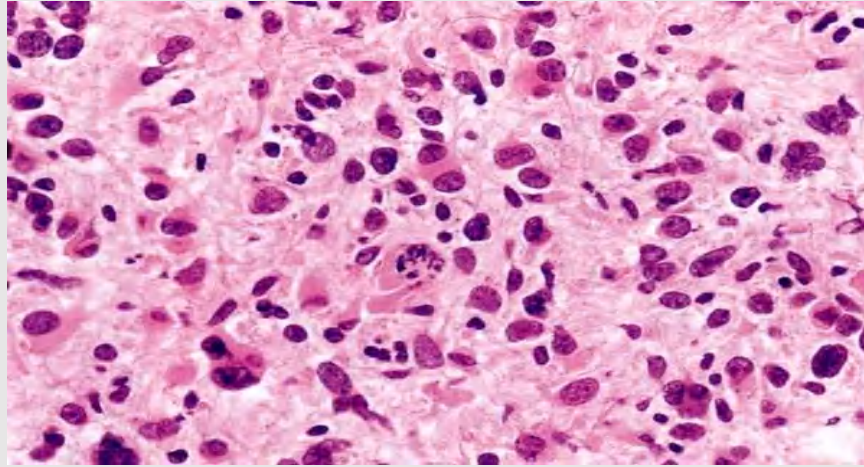
Medullary Carcinoma

- **Good prognosis.**
- groups of tumor cells with syncytial appearance (that is, seemingly fused cytoplasm, at left).
- There is typically also a lymphocytic and plasma cell infiltrate (right).



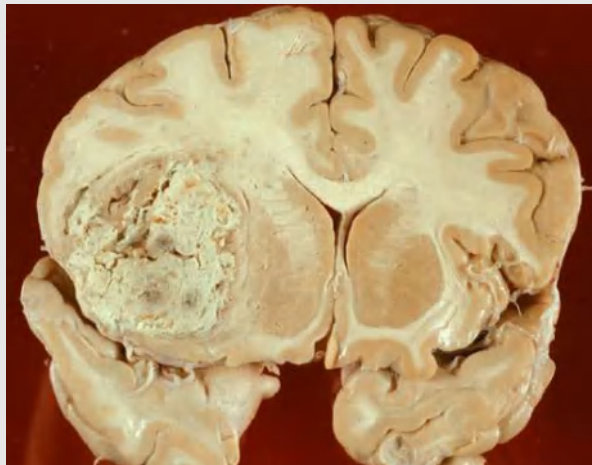
CNS

Anaplastic Astrocytoma



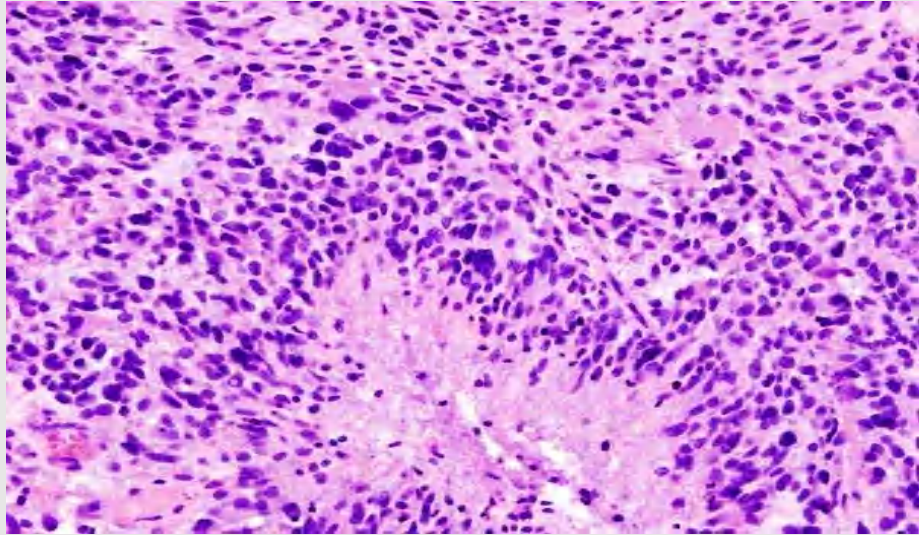
High cellularity, nuclear pleomorphism, frequent mitosis..

Astrocytoma glioblastoma multiform



Large mass, greyish, hemorrhage, necrosis, infiltration

Glioblastoma Astrocytoma



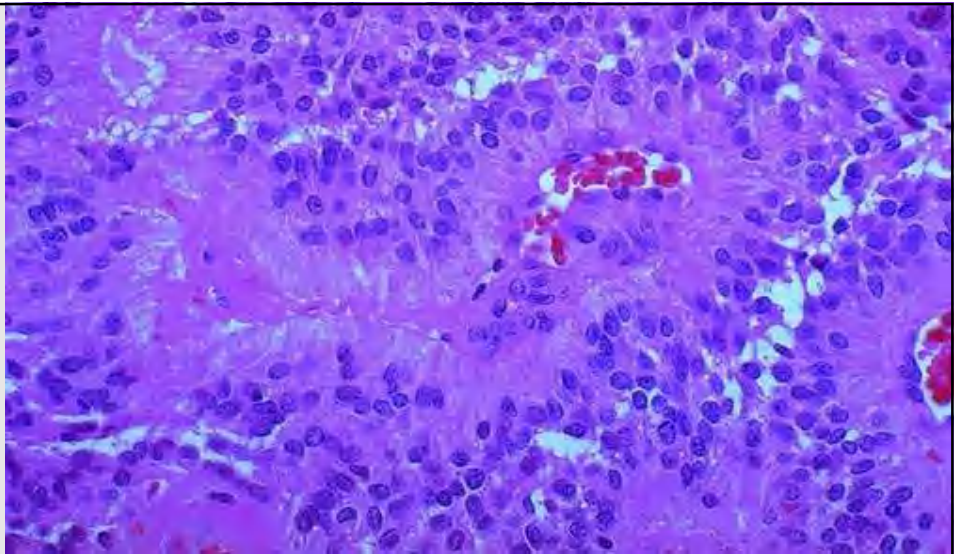
Small round cells, bizarre giant cells, area of necrosis with pseudopalisading (tumor cells surrounding a necrotic area), pleomorphic astrocytes, hemorrhage, necrosis.

Ependymoma



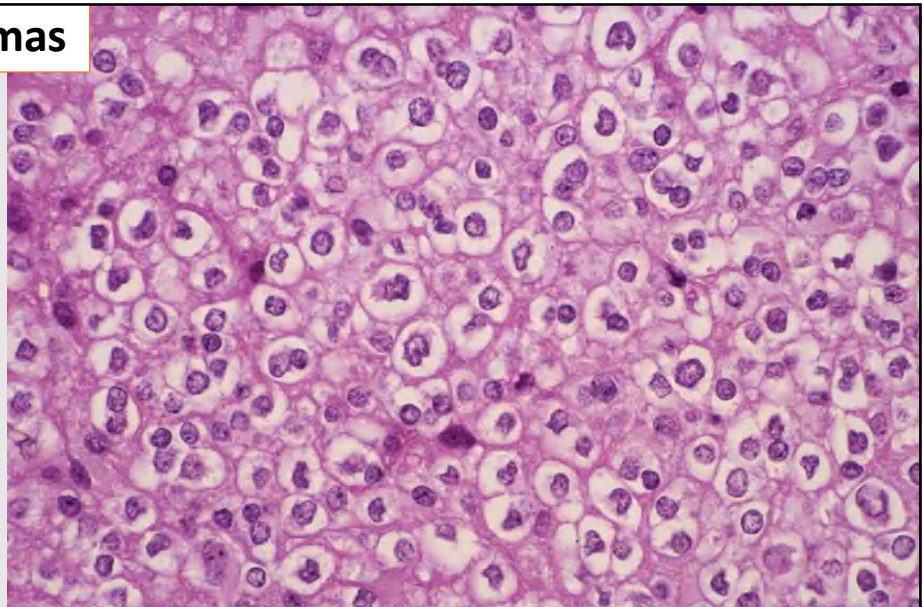
Solid, large, grayish mass arising from the ependymal lining of the fourth ventricle above the brainstem and bulging toward the cerebellum

Ependymoma



The tumor cells have regular, round to oval nuclei and abundant granular chromatin. Between the nuclei is a variably dense fibrillary background. Tumor cells may form round or elongated structures (rosettes, canals); more frequently present are perivascular pseudorosettes in which tumor cells are arranged around vessels.

Oligodendrogliomas



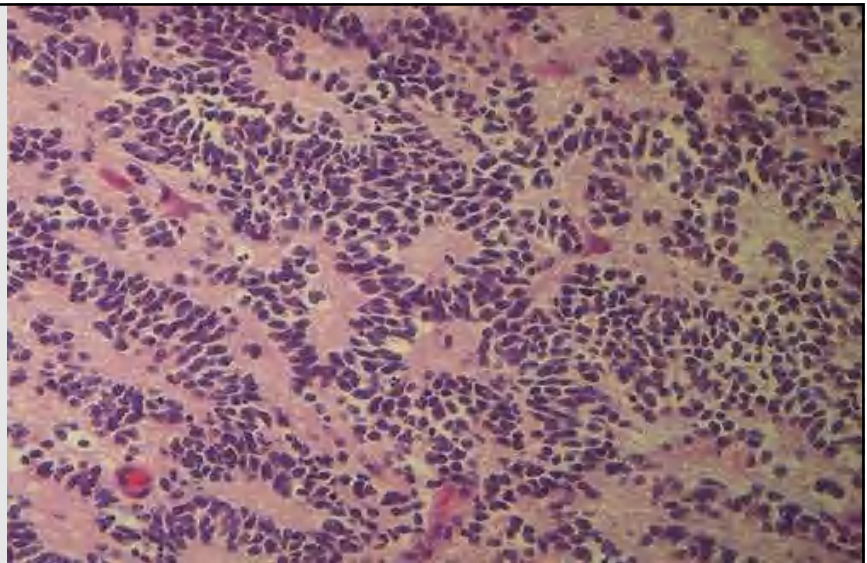
sheets of regular cells with spherical hyperchromatic nuclei surrounded by a clear halo of cytoplasm. Calcification is also seen

Medulloblastoma.

well circumscribed, irregular, grayish, friable mass near the midline of the cerebellum.



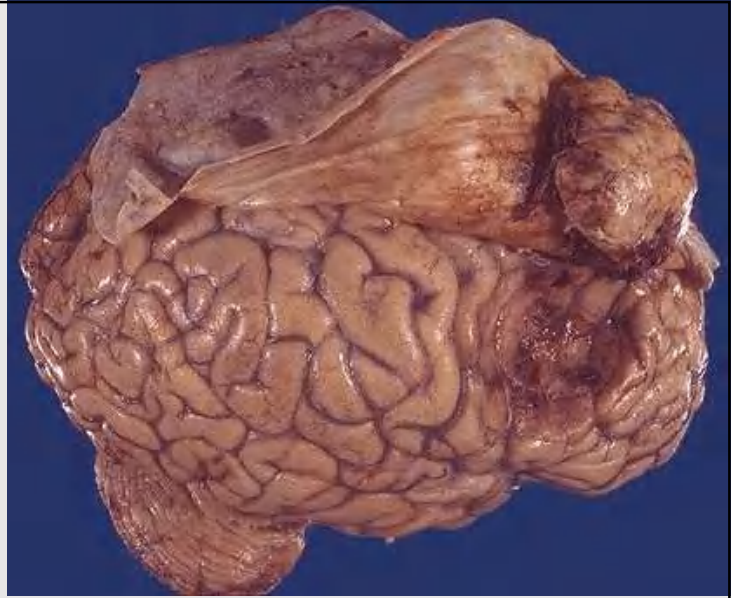
Medulloblastoma



Hypercellularity with small round blue cells with Homer-wright rosette (arrow)

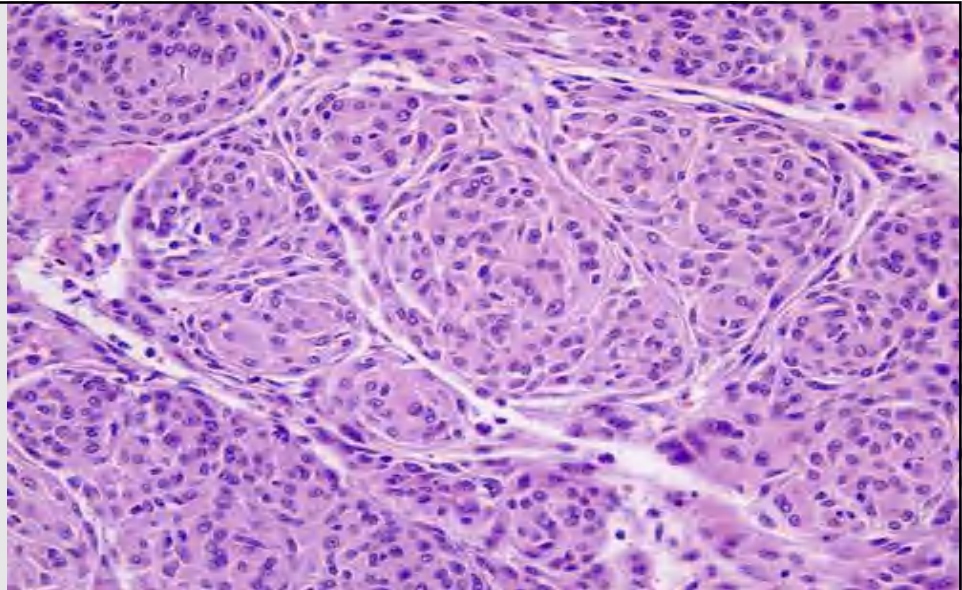
Meningioma

Solid, well circumscribed, rounded, (1-10 cm), grayish, areas of calcification, indent brain tissue.



Meningioma

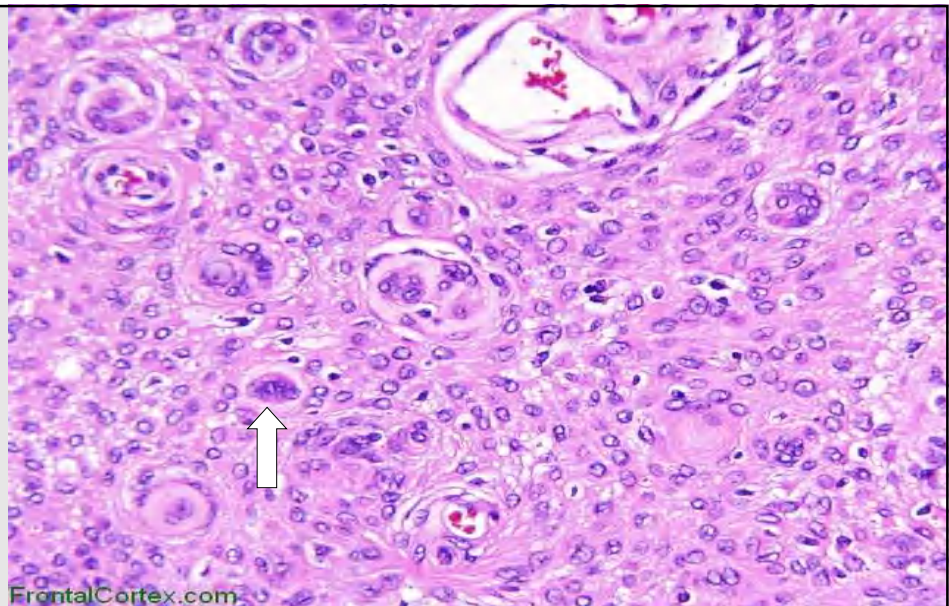
(Syncytial type)



Syncytial type: with whorls; nodules of meningiothelial cells, tight clusters of cells without visible cell membranes

Meningioma

(Transitional type)

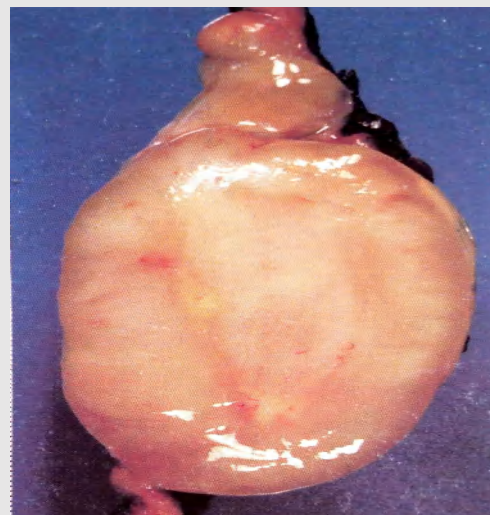


Features of the meningotheelial (whorls) and fibroblastic types, laminated calcification (psammomas bodies).

NEUROFIBROMA

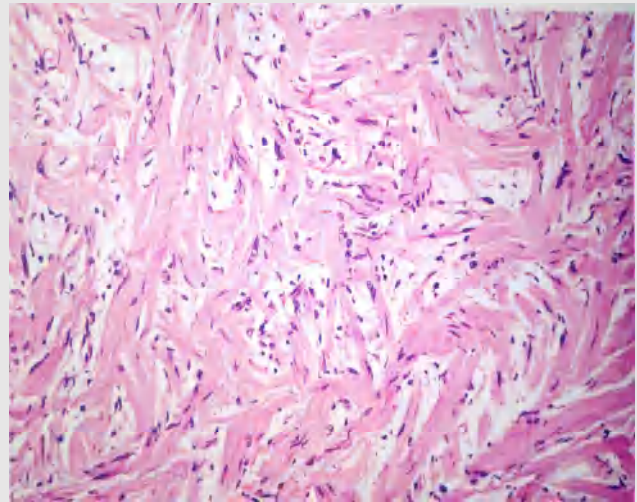
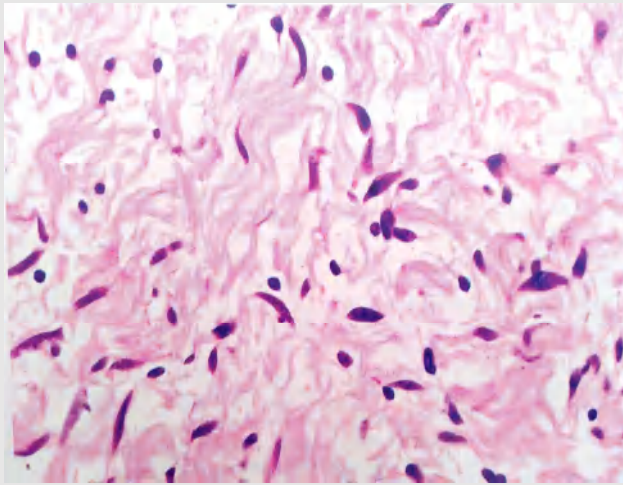


Multiple nodules, variant sizes



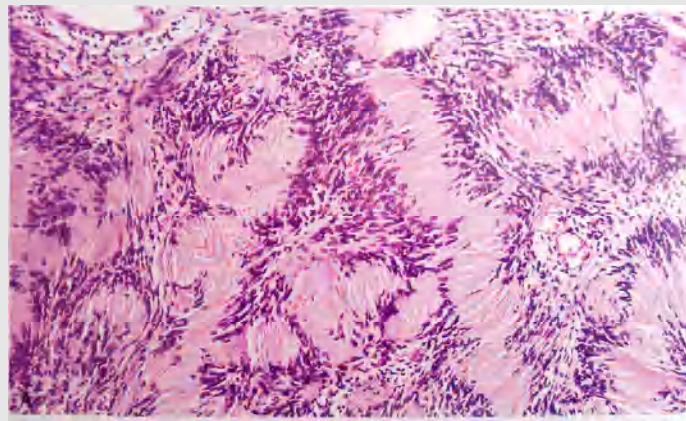
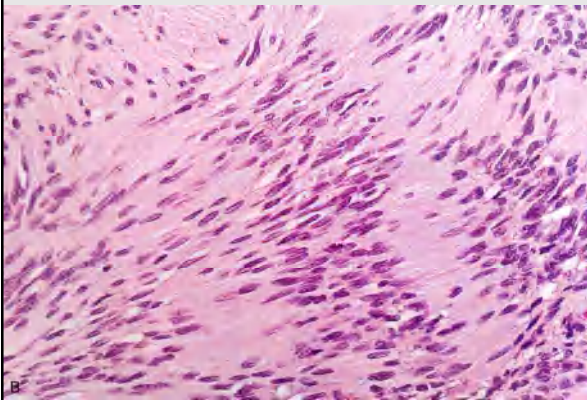
The tumors in the Schwann cells, myelin fiber and the axon itself.

NEUROFIBROMA



Schwann cells, fibrous tissue, proliferating axons, inflammatory cells

SCHWANNOMA



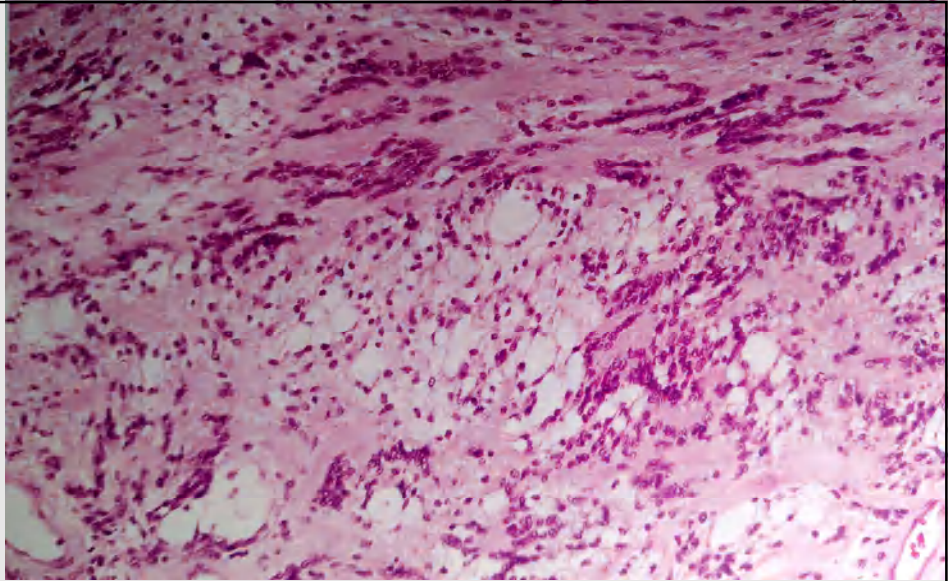
Antoni type A :

condensed cellular tumor, a uniform proliferation of neoplastic Schwann cells.

bland spindle cells with buckled nuclei are arranged into intersecting fascicles. These cells often align to produce nuclear palisading, resulting in alternating bands of nuclear and anuclear areas called Verocay bodies.

Axons are largely excluded from the tumor.

SCHWANNOMA



Antoni B:

Loose, hypocellular areas, the spindle cells are spread apart by a prominent myxoid extracellular matrix. Thick-walled hyalinized vessels often are present. Hemorrhage or cystic changes sometimes seen.

Blood and lymph

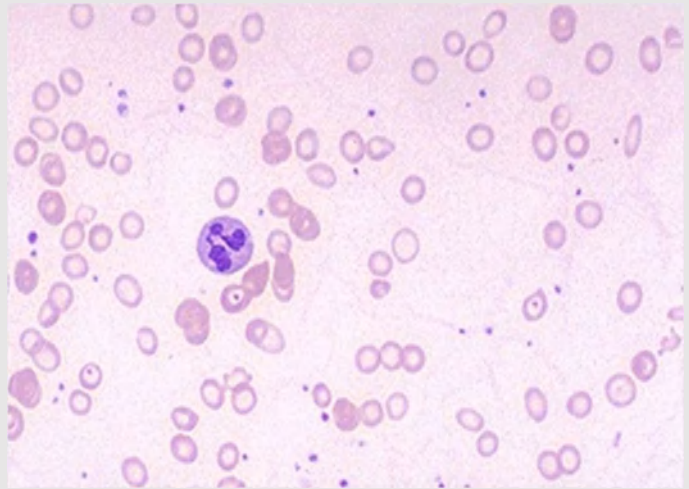
Iron deficiency anaemia



a. Koilonychia: typical spooning of nails.

Blood film shows:

1. The RBCs are smaller than normal and have an increased zone of central pallor.
2. Hypochromic (less hemoglobin in RBC).
3. Microcytic (smaller size of each RBC).
4. Anisocytosis (variation in size)
5. Poikilocytosis (variation in shape).
6. Target cells and pencil cells.

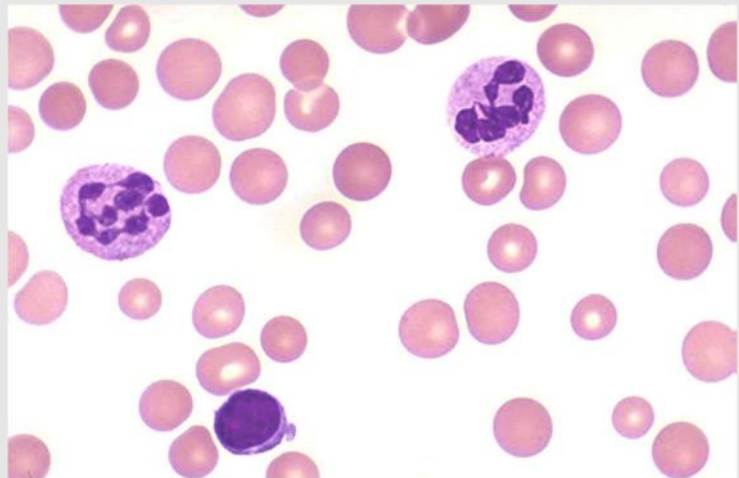


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Megaloblastic anaemia

Blood film shows:

1. Hypersegmented neutrophils.
2. Macro-ovalocytes (arrow): large oval RBCs lacking the central pallor.

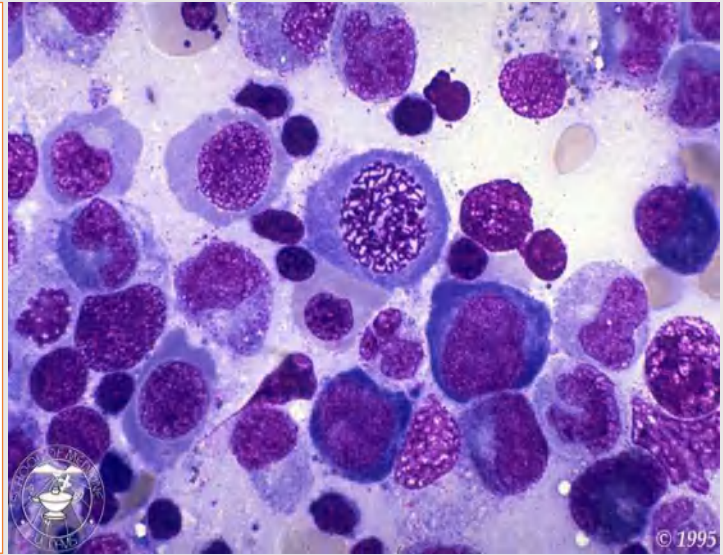


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Megaloblastic anaemia

Bone marrow shows:

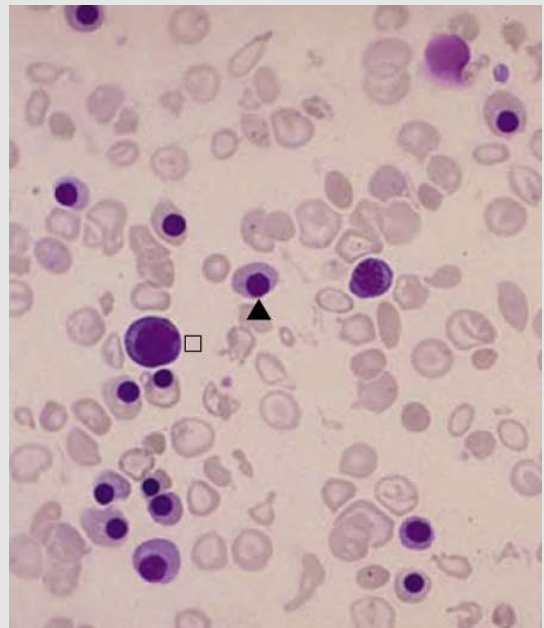
1. Hypercellularity.
2. Erythroblast: megaloblastic changes: (Asynchronous maturation)
 - Large cells.
 - Nucleus: lack of maturation, open sieve-like (مثل المشن) chromatin.
 - Cytoplasm: normal maturation.
3. Giant metamyelocytes.
4. Giant Band forms.



Thalassemia

α -Thalassemia major, Blood film:

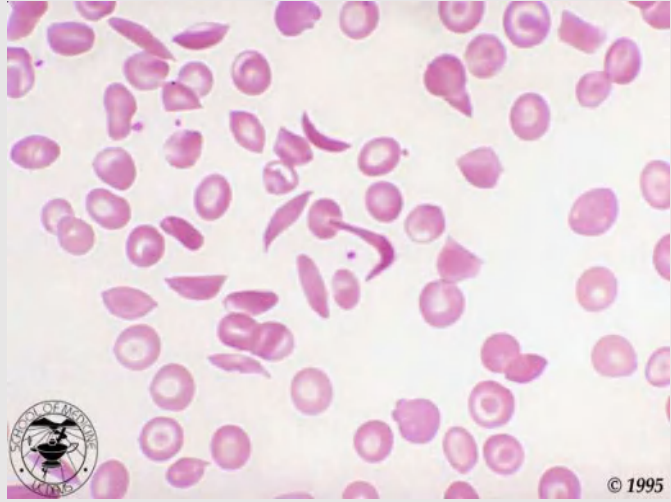
1. Anisocytosis (variation in size) of RBCs.
2. Poikilocytosis (variation in shape) of RBCs.
3. Expansion of erythropoiesis and presence of many immature RBCs in the peripheral blood, as evidenced by:
 - Nucleated RBCs (▲),
 - Erythroblasts (□).
4. Sever microcytic hypochromic.
5. Target cells.



Sickle cell anaemia



Necrosis and ulceration on medial Aspect of the ankle.



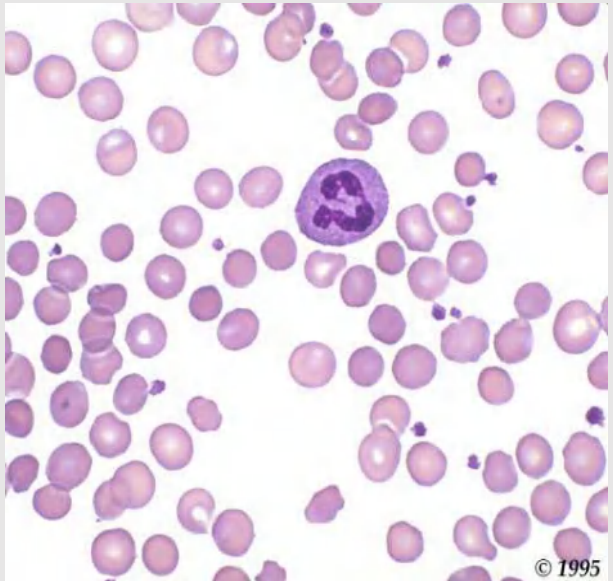
Blood film shows:

1. Some sickle cells.
2. Fragmented cells.
3. Target cells.
4. Nucleated cells.

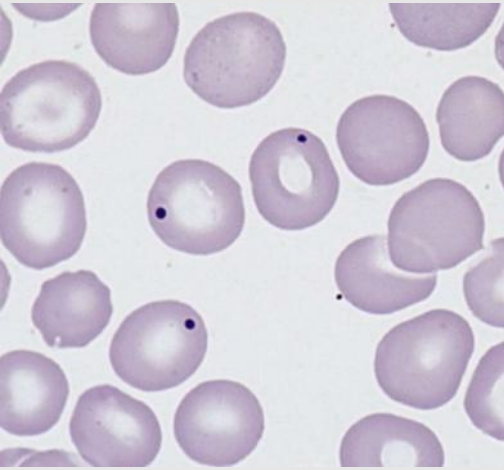
Hereditary spherocytosis

Blood film shows:

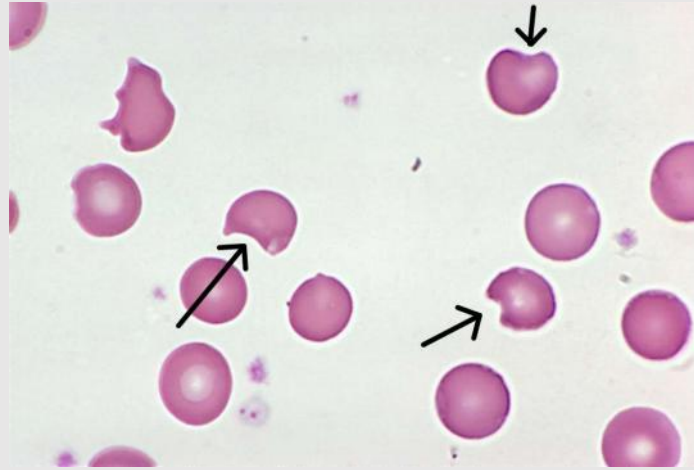
1. Micro-spherocytes: Small, Rounded, Hyperdense (lack of central pallor).
2. Occasional normoblast.



G6PD Deficiency



blood film shows:
Heinz bodies by **supravital stain**
(precipitated methemoglobin).



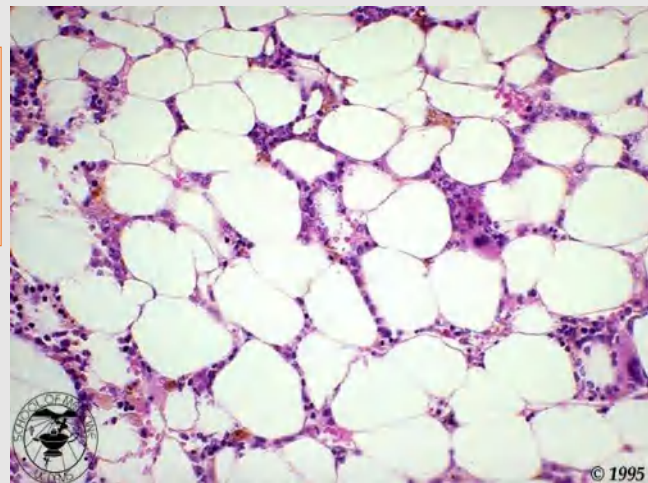
blood film shows:

- Fragmented and bite (blister) cells.

Aplastic anaemia

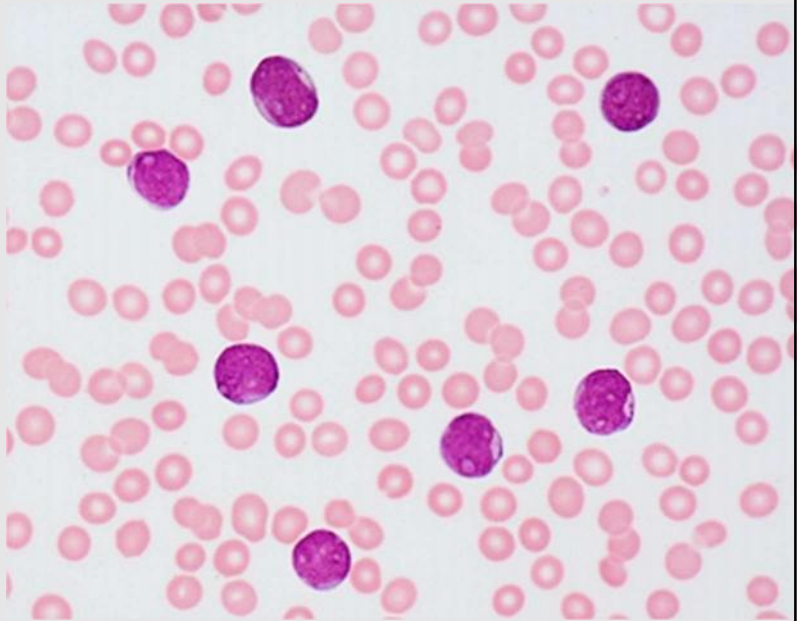
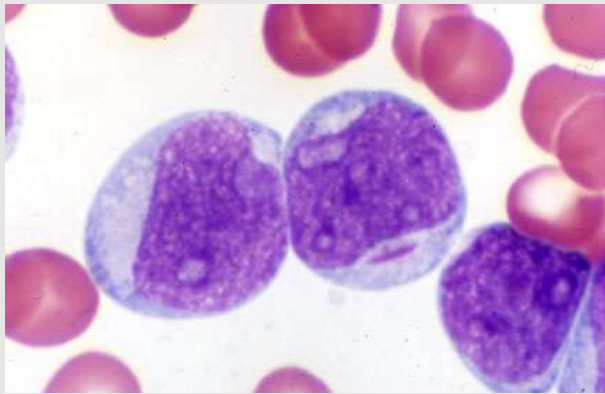
Bone marrow biopsy shows:

1. Hypocellularity.
2. Hematopoietic elements are markedly reduced.
3. Fatty replacement.

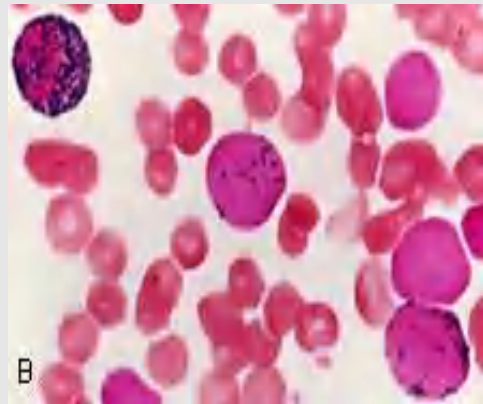


ALL**Blood Film:**

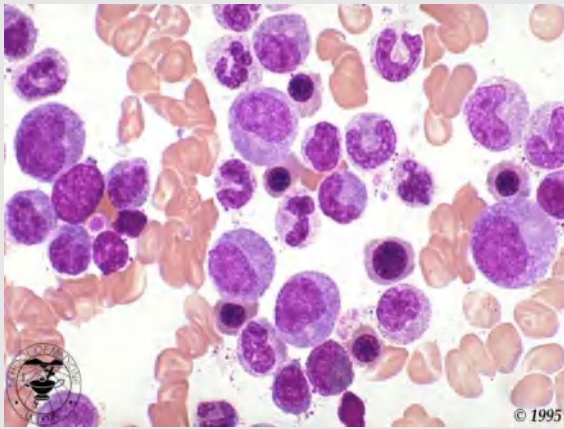
- Immature **Lymphoblasts** with scant cytoplasm and large nuclei that contain **nucleoli**.

**AML: M 1**

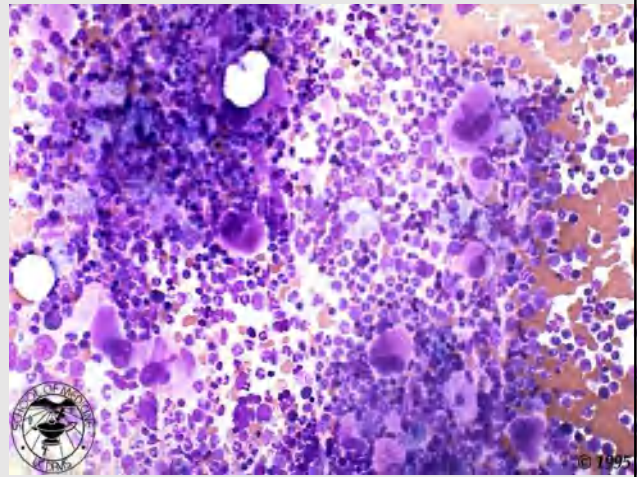
- Myeloblastic without maturation (Blasts > 90%).
- **Cytoplasm:** fine azurophilic **granulation**, **Auer rod**.
- **Nucleus:** one or more prominent nucleoli.

AML: M 2

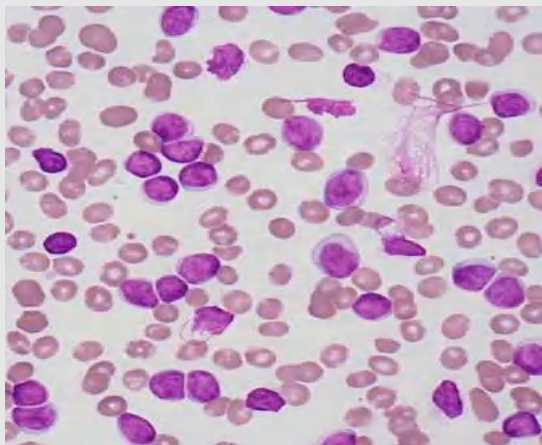
- $M_2 \rightarrow$ **Myeloblastic** with **granulocytic** maturation (Blasts 30% - 50%).
- **Peroxidase stain**
- Two leukemic myeloblasts with peroxidase positive granules.
- Intensely stained cell is a **more mature granulocyte**.
- Negatively stained cell could be lymphocyte.

CML**Blood Film:**

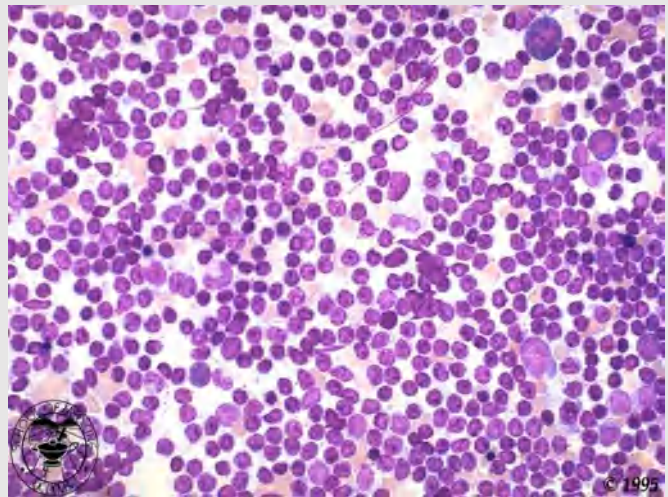
- Leukocytosis (mainly: basophils, eosinophils and bands).
- Complete spectrum of myeloid series in blood with predominance of **myelocytes** and **neutrophils**.

**Bone marrow examination:**

- Hypercellular B.M
- Predominance of **myelocytes**.

CLL**Blood film:**

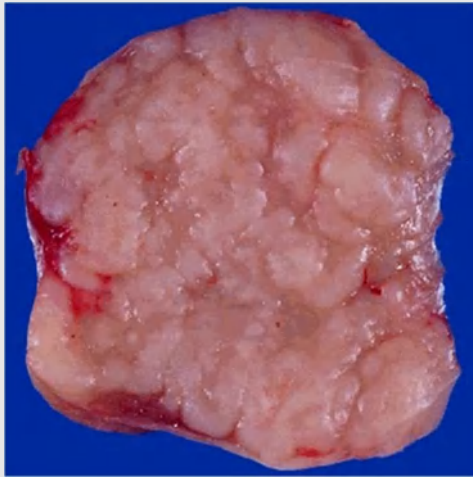
- **Lymphocytosis:**
- Small, normally appearing mature lymphocytes.
- Smudge (smear) cells are present.

**Bone marrow:**

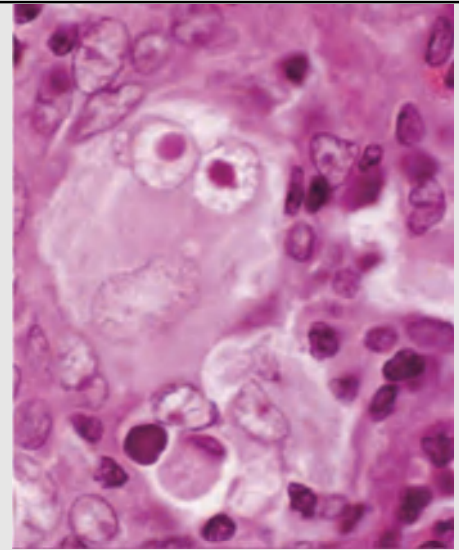
- Hypercellular.
- Replacement of B.M elements by lymphocytes.

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Hodgkin's lymphoma

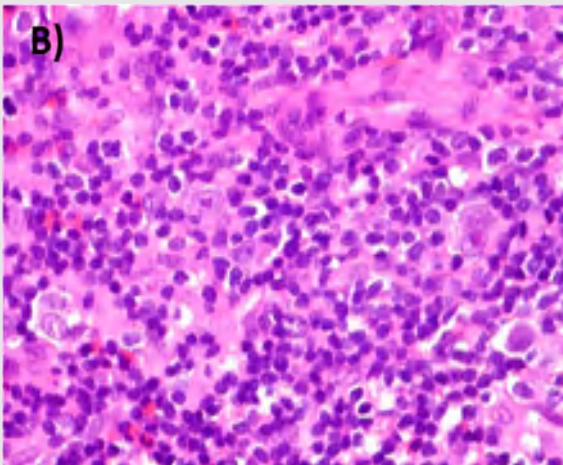


enlarged (> 1cm) node, firm, irregular, lobulated.
No or minimal necrosis and hemorrhage.
Cut-section: grayish pink, homogenous.



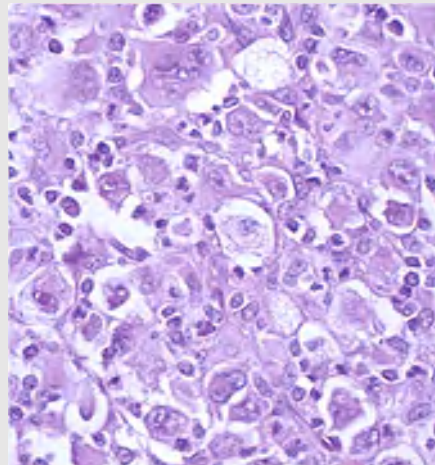
Reed-Sternberg cell: 2 mirror image nuclei (owl-eyed) surrounded by clear zone. Eosinophilic nucleolus & abundant eosinophilic cytoplasm.
Non-neoplastic cells: mixture of Leukocytes.

Hodgkin's lymphoma: Mixed cellularity



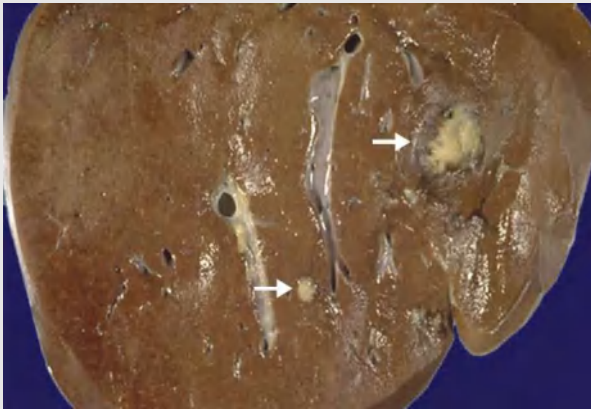
B. mixed population of lymphocytes, plasma cells, eosinophils, histiocytes and Reed-Sternberg cells.
(Stain: Hematoxylin & eosin).

Hodgkin's lymphoma: Lymphocyte depletion

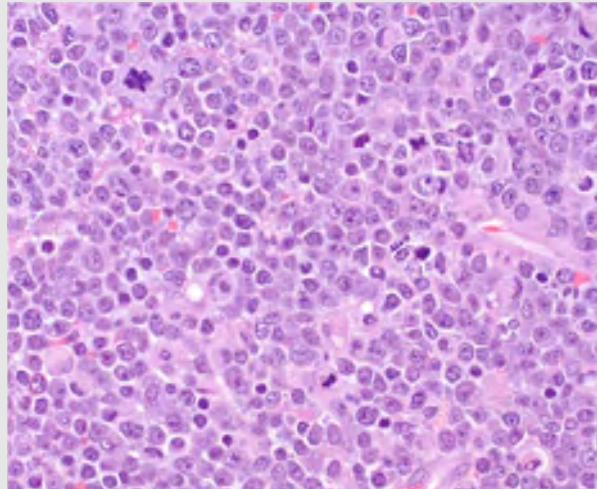


- **Many** Reed-Sternberg variants.
- **Few** lymphocytes.
- Some association with **HIV infection**.
- It is often associated with **EBV infection**.

Non- Hodgkin's lymphoma: Diffuse large B-cell lymphoma

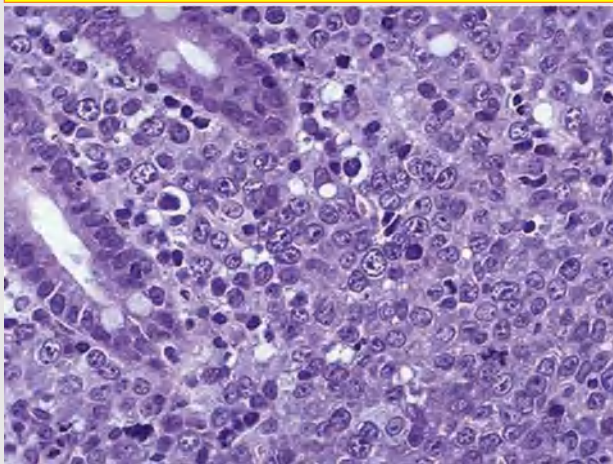


- Seen here on cut surface of liver are two rounded pale tan mass lesions (**arrows**).
- The color can range from white to tan to red, often **intermixed**.



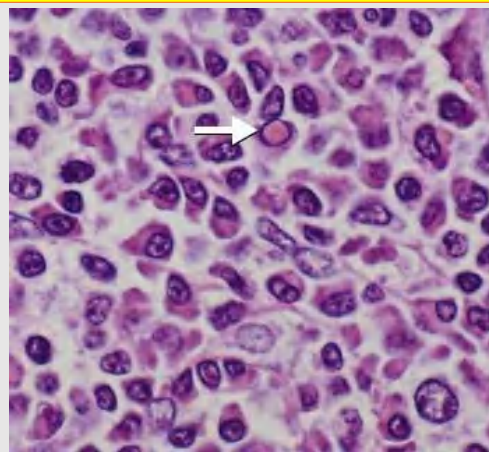
- Large cells with large nuclei & prominent nucleoli.
- Mitoses are frequent.
- The cells mark with CD10, CD19, and **CD20**.

NHL: Burkitt lymphoma



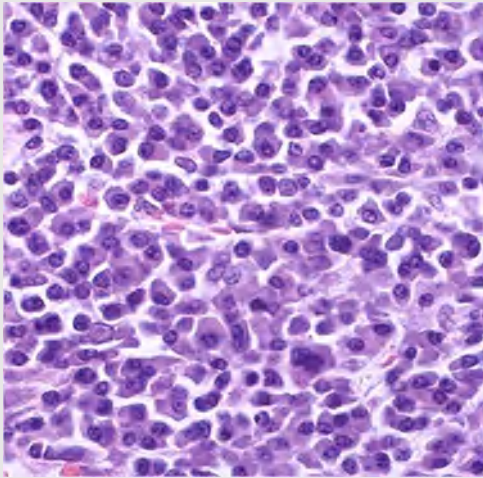
- Small intestinal mucosa with large infiltrating cells of Burkitt lymphoma (BL).
- Mitoses and apoptosis with cellular debris cleared by large **macrophages** producing a "**starry sky**" pattern are prominent features.

NHL: Lymphoplasmacytic lymphoma



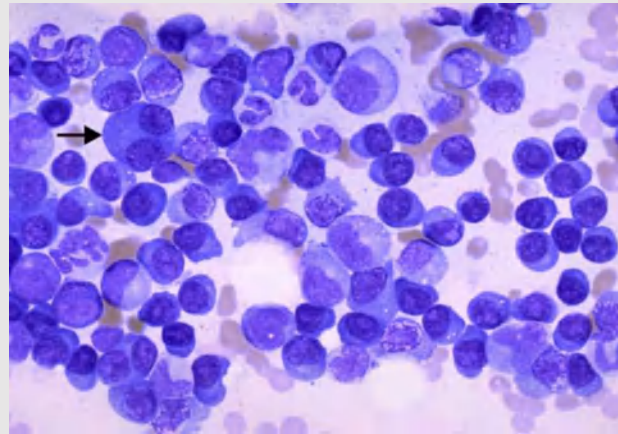
- **Small lymphocytes** with **plasmacytoid cells** and few **large cells**.
- Characteristic **PAS-positive** intranuclear inclusions (**arrows**) known as **Dutcher bodies**.

Multiple myeloma



Bone marrow biopsy:

- Sheets of plasma cells that are very similar to normal plasma cells.



A smear of bone marrow aspirate:

- Plasma cells constituting more than 30% of cellularity.
- Numerous plasma cells with a **perinuclear halo of clearer cytoplasm** (Golgi apparatus).
- Larger abnormal plasma cells, one double-nucleated (**arrows**).
- Clear cytoplasmic droplets contain immunoglobulin.

Thank you..

Dr. Hamzah AD